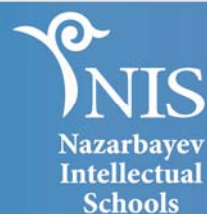


AEO «Nazarbayev intellectual schools»
Center of pedagogical measurements



**EXAMPLE CANDIDATE
RESPONSE OF EXTERNAL
SUMMATIVE ASSESSMENT**

12 grade

CHEMISTRY
Paper 2

Astana, 2023 year

Contents

QUESTION 1(a)(i), (ii)	3
QUESTION 1(b) (i), (ii)	7
QUESTION 1(c) (i), (ii), (iii), (iv)	10
QUESTION 2(a) (i), (ii)	14
QUESTION 2(b)(i), (ii)	17
QUESTION 3(a)(i), (ii)	21
QUESTION 3(b)	24
QUESTION 3(c)	27
QUESTION 4(a) (i), (ii)	29
QUESTION 4(b), (c) (i), (ii)	33
QUESTION 4(d)	37
QUESTION 5 (a), (b)	40
QUESTION 5 (c), (d)	44
QUESTION 6(a) (i), (ii), (iii)	47
QUESTION 6(b), (c)	51
QUESTION 7(a)(i), (ii)	54
QUESTION 7(b)(i), (ii) (c)	58
QUESTION 7(d)	62
QUESTION 8(a) (b) (c) (i), (ii)	64
QUESTION 8(d) (i), (ii)	68
QUESTION 9(a)(i), (ii), (iii)	70
QUESTION 9(b) (i), (ii)	73
QUESTION 9(d)	77
QUESTION 10(a)	79
QUESTION 10 (b) (i), (ii), (iii)	81
QUESTION 11 (a) (b) (i), (ii)	84
QUESTION 11(c) (d)	86

QUESTION 1(a)(i), (ii)

1 Sulfur hexafluoride, SF₆, is used in lung radiography.

For
Examiners
Use

- (a) (i) Draw the dot-and-cross diagram of the sulfur hexafluoride, SF₆ molecule (show only the outer shell electrons).

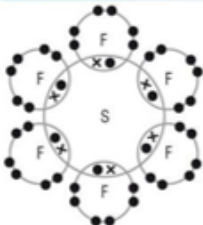
[2]

- (ii) Predict shape and bond angles of sulfur hexafluoride.

shape [1]

bond angle [1]

MARK SCHEME

Question	Answer	Mark	Additional guidance
1 (a) (i)		2	M1 for showing 12 electrons around <u>sulfur</u> and 8 electrons of fluorine atoms M1 for correct dot-and-cross electrons
1 (a) (ii)	octahedral and 90° all angles	2	M1 for the shape M2 mark for the correct angle

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

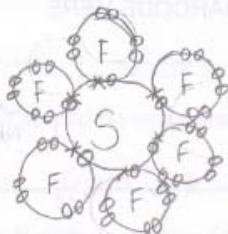
1 Sulfur hexafluoride, SF_6 , is used in lung radiography.

- (a) (i) Draw the dot-and-cross diagram of the sulfur hexafluoride, SF_6 molecule (show only the outer shell electrons).

SF_6 - covalent

oo - outer electrons of F.

xx - outer electrons of S.



[2]

2/2

- (ii) Predict shape and bond angles of sulfur hexafluoride.

shape octahedral ✓

[1]

1/1

bond angle 90° ✓

[1]

1/1

EXAMINER'S COMMENTS

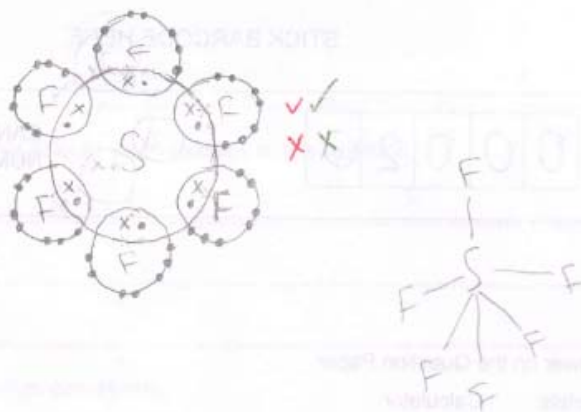
This candidate received four marks. A student clearly demonstrated all the electrons in fluorine and the electron pairs between F and S. The student correctly found the shape and bond angle of sulfur hexafluoride.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

1 Sulfur hexafluoride, SF_6 , is used in lung radiography.

- (a) (i) Draw the dot-and-cross diagram of the sulfur hexafluoride, SF_6 molecule (show only the outer shell electrons).

For
Examiner's
Use



[2]

1 1

- (ii) Predict shape and bond angles of sulfur hexafluoride.

shape octahedral ✓ ✓

[1]

1 1

bond angle 90° ✓

[1]

0 0

EXAMINER'S COMMENTS

This candidate received two marks.

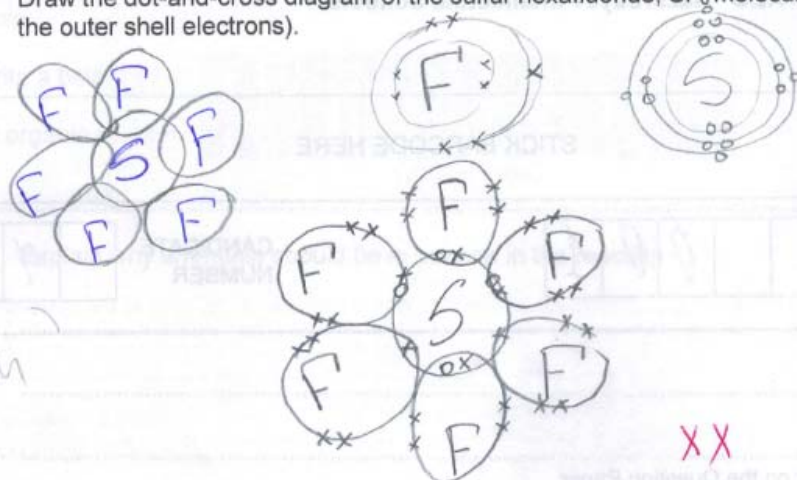
In (i) question, the candidate correctly showed the common electron pair, but placed the electron incorrectly on fluorine.

In question (ii), the candidate provided the shape but missed the bond angle.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

1 Sulfur hexafluoride, SF_6 , is used in lung radiography.

(a) (i) Draw the dot-and-cross diagram of the sulfur hexafluoride, SF_6 , molecule (show only the outer shell electrons).



(ii) Predict shape and bond angles of sulfur hexafluoride.

shape octahedral ✓ [1] 1
bond angle 60° X [1] 0

For
Examiner's
Use

EXAMINER'S COMMENTS

This candidate received only one mark.

In question (i), the candidate incorrectly shows the shared electron pairs and the electrons on fluorine.

In question (ii), the candidate correctly defines the octahedral shape, but the bond angle is incorrect.

QUESTION 1(b) (i), (ii)

(b) (i) The bonding in a sulfur hexafluoride, SF₆ molecule is covalent.

Explain the meaning of a covalent bond by using the SF₆ molecule.

.....
..... [1]

(ii) The covalent bonds between atoms in the SF₆ molecule are covalent *polar* bonds.

Explain why the molecule is non-polar.

.....
..... [1]

MARK SCHEME

(b) (i)	Covalent bond is pair of electrons shared between two atoms	1	Sharing is the key <u>word</u>
(b) (ii)	the six polar bonds are symmetrically arranged /dipoles cancel each other out	1	

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

(b) (i) The bonding in a sulfur hexafluoride, SF_6 molecule is covalent.

Explain the meaning of a covalent bond by using the SF_6 molecule.

Covalent bond is a bonding type between non-metals by sharing
outer electrons. [1]

(ii) The covalent bonds between atoms in the SF_6 molecule are covalent polar bonds.

Explain why the molecule is non-polar.

The S-F is a polar, but total molecule is non-polar due to
the total symmetry (no dipole moment) [1]

EXAMINER'S COMMENTS

In question (b), this candidate received two marks for the correct use of key words: sharing electrons in (i) and symmetry in (ii). The student also wrote in (ii) "no dipole moment", demonstrating a correct understanding.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

(b) (i) The bonding in a sulfur hexafluoride, SF_6 molecule is covalent.

Explain the meaning of a covalent bond by using the SF_6 molecule.

bond in which each molecule ^(molecule) contributes electrons to use them on
their orbitals simultaneously and thus bond [1]

(ii) The covalent bonds between atoms in the SF_6 molecule are covalent polar bonds.

Explain why the molecule is non-polar.

because it is symmetrical, thus polarities
cancel out and yield in non-polar molecule [1]

EXAMINER'S COMMENTS

In question (b), only 1 mark was awarded for the correct explanation with key words in (ii).

In question (i), the candidate wrote a description without using key words.

SAMPLE OF CANDIDATE RESPONSE GRADED “E”

- (b) (i) The bonding in a sulfur hexafluoride, SF_6 molecule is covalent.

Explain the meaning of a covalent bond by using the SF_6 molecule.

Covalent bond is a bond between two or more molecules
of non-metals such as in the SF_6 X [1]

- (ii) The covalent bonds between atoms in the SF_6 molecule are covalent polar bonds.

Explain why the molecule is non-polar.

Because they are both non-metals X [1]

EXAMINER'S COMMENTS

In question (b), the candidate does not receive any marks. In question (i), the candidate incorrectly describes the concept of two nonmetals instead of discussing shared electron pairs.

In question (ii), the candidate fails to explain why the molecule is nonpolar and instead attempts to explain the bonds within.

QUESTION 1(c) (i), (ii), (iii), (iv)

(c) Many metal ions form complex ions, which have the same shape as a SF_6 molecule. The complex ion, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is formed when water ligands surround Co^{2+} ion.

(i) Explain the meaning of the term *ligand*.

.....
 [1]

(ii) Name the type of bond formed between the water ligands and Co^{2+} .

..... [1]

(iii) Cobalt chloride paper is used in the laboratory to test the water. Sky blue cobalt chloride paper contains the complex ion $[\text{Co}(\text{Cl}_4)]^{2-}$, when water is added the pink coloured $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ complex is formed.

Explain why the colour changes.

.....
 [1]

(iv) Write the equation for the formation of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ from the cobalt chloride paper.

Name this type of reaction.

equation [1]

type of reaction [1]

[Total: 11]

MARK SCHEME

(c) (i)	a species that donates a lone pair of electrons to the metal cation or central atom	1	Accept only the explanation with words: lone pair donor
(c) (ii)	dative covalent bonds/co-ordinate bonds	1	
(c) (iii)	different ligand affects to the color / another coordination number / the size of the energy gap between 3d orbitals is different	1	
(c) (iv)	$[\text{Co}(\text{Cl}_4)]^{2-} + 6\text{H}_2\text{O} \leftrightarrow [\text{Co}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^-$ Ligand exchange / ligand substitution reaction	1 1	Accept without word 'ligand'

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

c) Many metal ions form complex ions, which have the same shape as a SF_6 molecule. The complex ion, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is formed when water ligands surround Co^{2+} ion.

(i) Explain the meaning of the term *ligand*.

ligand is a donor of lone pairs

[1]

(ii) Name the type of bond formed between the water ligands and Co^{2+} .

co-ordination covalent bond

[1]

(iii) Cobalt chloride paper is used in the laboratory to test the water. Sky blue cobalt chloride paper contains the complex ion $[\text{Co}(\text{Cl}_4)]^{2-}$, when water is added the pink coloured $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ complex is formed.

Explain why the colour changes.

because the ligand's identity and coordination ~~for~~ number are changed affecting the amount of energy absorbed by electrons from visible light spectrum

[1]

(iv) Write the equation for the formation of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ from the cobalt chloride paper.

Name this type of reaction.

equation $[\text{CoCl}_4]^{2-} + 6\text{H}_2\text{O} \rightarrow [\text{Co}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^-$

[1]

type of reaction ligand substitution reaction

[1]

For
Examiner's
Use

EXAMINER'S COMMENTS

In question (c), this candidate received full marks. In question (i), the candidate correctly defines the meaning of ligand using key words. In question (ii), the candidate defines the type of bond.

In question (iii), the candidate accurately states that the colour changes as the coordination number changes. In question (iv), the candidate correctly writes a balanced equation and identifies the type of reaction.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

(c) Many metal ions form complex ions, which have the same shape as a SF_6 molecule. The complex ion, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is formed when water ligands surround Co^{2+} ion.

(i) Explain the meaning of the term *ligand*.

a molecule which can donate and
bond via electron pair [1] X 0.
(x2 covalent or some
sort)

(ii) Name the type of bond formed between the water ligands and Co^{2+} .

dative covalent ✓ [1]

For
Examiner's
Use
1

(iii) Cobalt chloride paper is used in the laboratory to test the water. Sky blue cobalt chloride paper contains the complex ion $[\text{Co}(\text{Cl}_4)]^{2-}$, when water is added the pink coloured $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ complex is formed.

Explain why the colour changes.

because of the different molecular
structures, compounds absorb different
light spectrums, thus changing reflected color occurs [1] X 0

(iv) Write the equation for the formation of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ from the cobalt chloride paper.

Name this type of reaction.

equation $[\text{Co}(\text{Cl}_4)]^{2-} + 6\text{H}_2\text{O} \rightarrow [\text{Co}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^-$ ✓ [1] 1

type of reaction ligand substitution ✓ [1] 1

EXAMINER'S COMMENTS

In this question, the candidate received only two marks.

The answer in (c) (i) does not contain key words "donate lone pair".

In question (c) (ii), the candidate correctly identifies the type of bond between the ligand and the central atom.

In question (c) (iii), the candidate attempts to explain why the colour changes, but it is not precise. In question (c) (iv), the candidate correctly writes a balanced equation and names the type of reaction.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

(c) Many metal ions form complex ions, which have the same shape as a SF_6 molecule. The complex ion, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is formed when water ligands surround Co^{2+} ion.

(i) Explain the meaning of the term *ligand*.

ligand is molecule or ion with at least 1 or more
co-ordination numbers, which forms dative bonds or bonds [1]

(ii) Name the type of bond formed between the water ligands and Co^{2+} .

dative bond [1]

(iii) Cobalt chloride paper is used in the laboratory to test the water. Sky blue cobalt chloride paper contains the complex ion $[\text{Co}(\text{Cl}_4)]^{2-}$, when water is added the pink coloured $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ complex is formed.

Explain why the colour changes.

Because it ~~absorb~~ the molecule absorbs energy / light, and d-orbitals
split into 2 energetical levels and the electrons jump from
low to high [1]

(iv) Write the equation for the formation of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ from the cobalt chloride paper.

Name this type of reaction.

equation $[\text{Co}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^- \rightarrow [\text{CoCl}_4]^{2-} + 6\text{H}_2\text{O}$ [1]

type of reaction substitution [1]

For
Examiner's
Use

EXAMINER'S COMMENTS

In this question, the candidate received only two marks.

In question (c) (i), the answer does not contain the key words "donate lone pair".

In question(c) (ii), the candidate correctly identifies the type of bond between the ligand and the central atom.

In question(c) (iii), the candidate attempts to describe why this compound has a colour but does not answer, why the colour changes.

In question (c) (iv), the candidate writes the equation in the incorrect direction but correctly names the type of this reaction.

QUESTION 2(a) (i), (ii)

2 The trends in the Periodic Table help explain the properties of a particular chemical element.

(a) (i) Define the term for the first ionisation energy.

.....

 [2]

(ii) Explain why the first ionisation energy of potassium is smaller than the first ionisation energy of bromine.

.....

 [2]

MARK SCHEME

2 (a) (i)	the energy required to remove one mole of electron from one mole of gaseous atoms	1 1	Accept if student wrote correct equation for any element [1] It is important to include state symbols [1]
2 (a) (ii)	Potassium has larger atomic radius (greater distance between nucleus and outermost electron) / less number of protons in the nucleus/smaller nuclear charge less attraction between nucleus and the outermost electron	1 1	Accept if a student gives an answer with respect to bromine

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

2 The trends in the Periodic Table help explain the properties of a particular chemical element.

(a) (i) Define the term for the first ionisation energy.

First ionisation energy is the energy required to remove one mole of electrons from one mole of atoms of an element to form +1 gaseous ions. [2]

(ii) Explain why the first ionisation energy of potassium is smaller than the first ionisation energy of bromine.

${}_{19}\text{K} < {}_{35}\text{Br}$ *Measure energy*
The ^{first} ionisation energy of potassium is smaller because the nuclear charge of bromine is larger, outer electrons more attracted to the nucleus (atomic radius decrease). [2]

For
Examiner's
Use

2 2

2 2

EXAMINER'S COMMENTS

This candidate received all four marks.

In question (i), the candidate used all the necessary key words.

In question (ii), the candidate explained the question by comparing nuclear charging and the attraction between the nucleus and outer electrons.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

2 The trends in the Periodic Table help explain the properties of a particular chemical element.

(a) (i) Define the term for the first ionisation energy.

It is the energy needed to remove 1 e^- from
each molecule of 1 mole of this element under standard
conditions with everything in its standard state [2] ~~X~~ 0

(ii) Explain why the first ionisation energy of potassium is smaller than the first ionisation energy of bromine.

because Br has more ~~✓~~ protons on the
same energy level, thus stronger
attraction ~~✓~~ forces between 'p' and e^-
and smaller ~~✓~~ orbital radius, thus more
energy needed for e^- to escape from Br than K [2] 2

EXAMINER'S COMMENTS

This candidate received only two marks.

In (i), the candidate missed the key word 'one mole' and incorrectly wrote 'elements' instead of 'one mole of gaseous atoms'.

In (ii), the candidate correctly used the key words 'more protons' and 'stronger attraction'.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

2 The trends in the Periodic Table help explain the properties of a particular chemical element.

(a) (i) Define the term for the first ionisation energy.

It is the energy required for the electrons to give. When the
electrons are pushed the ion forms, which has more electronegativity
[2] ~~XX~~ 0

(ii) Explain why the first ionisation energy of potassium is smaller than the first ionisation energy of bromine.

Because Br has more atomic number and bigger radius ~~✓~~ also
there are more outer shells in bromine than in the potassium.
So bromine requires more energy
[2] ~~X~~ 0

EXAMINER'S COMMENTS

This candidate didn't receive any marks because he didn't demonstrate what first ionization energy means and how it changes.

QUESTION 2(b)(i), (ii)

- (b) (i)** A potassium ion, K^+ , and a calcium ion, Ca^{2+} , both have the same number of electrons.

Explain why the ionic radius of potassium ion is larger.

.....
 [1]

- (ii)** Describe the change in *atomic radii* across Period 3.

Explain your answer.

.....

 [3]

MARK SCHEME

2 (b) (i)	less nuclear charge of potassium ion / less number of protons in potassium ion/ less atomic number of potassium	1	do NOT accept less/greater distance between nucleus and outermost electron
2 (b) (ii)	Atomic radius decreases across the period, because: the positive charge of the nucleus increases/number of protons/atomic number increases across the period the force of attraction between the nucleus and electrons increases	1 1 1	

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

- (b) (i) A potassium ion, K^+ , and a calcium ion, Ca^{2+} , both have the same number of electrons. $_{19}K^+ 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ $Ca^{2+} 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

Explain why the ionic radius of potassium ion is larger.

The nuclear charge of Ca^{2+} is more positive and outer electrons more attracted to the nucleus which results in small decrease of atomic radius. [1]

- (ii) Describe the change in atomic radii across Period 3.

Explain your answer.

The atomic radius decrease across the Period 3. Because they all have same number of orbitals, but different number of electrons and nuclear charge. The increasing number of electrons and nuclear charge means that electrons are more attracted to the nucleus.

[3]

[Total: 8]

1 1

3/3

8 8

EXAMINER'S COMMENTS

In question (b)(i), the candidate received one mark because he wrote that the ionic radius of Ca^{2+} is larger, stating that it is due to the presence of more nuclear charge.

In question (b)(ii), the candidate received all three marks because they correctly described the changes in atomic radii and explained it using the concept of nuclear charge and the attraction between the nucleus and outer electrons.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

- (b) (i) A potassium ion, K^+ , and a calcium ion, Ca^{2+} , both have the same number of electrons.

Explain why the ionic radius of potassium ion is larger.

K^+ has 19 e^- while Ca^{2+} has 20, thus more $\checkmark$ charge in the nucleus and stronger [1]
attraction forces on same number of e^-

- (ii) Describe the change in atomic radii across Period 3.

Explain your answer.

generally it decreases $\checkmark$ along the period
as the charge density increases and
orbital amount stays in limits of 3rd
level, thus along the way $F \uparrow$ (between $p \cdot e^-$)
thus $\downarrow R$ as attraction forces attract
the particles.

[3]

EXAMINER'S COMMENTS

In question (b)(i), the candidate received one mark because he wrote that the ionic radius of Ca^{2+} is larger due to the increased nuclear charge in Ca^{2+} .

In question (b)(ii), the candidate received one mark for describing the changes in atomic radii and one mark for explaining it based on changes in nuclear charge.

However, they lost one mark for providing unclear explanations regarding the attraction between the nucleus and electrons.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

- (b) (i) A potassium ion, K^+ , and a calcium ion, Ca^{2+} , both have the same number of electrons.

Explain why the ionic radius of potassium ion is larger.

Because it has ^{larger} more distance between the nuclei and electrons.

..... X [1] 0

- (ii) Describe the change in atomic radii across Period 3.

Explain your answer.

the atomic radii increases ^X to the right across the period 3
and increases down the group. Because across the period the
atomic number increases to the right, and the number of
outer shells also increases, which increases the ~~shielding~~ distance between
nuclei and electrons, also the shielding effect and electronegativity
increases. XX

..... [3] 0

EXAMINER'S COMMENTS

The candidate receives no marks.

In question (b)(i), the candidate repeated the statement about the ionic radius being greater in the calcium ion but did not explain why.

In question (b)(ii), the candidate stated an incorrect trend in atomic radii and attempted to explain it by considering the number of outer shells and the shielding effect.

QUESTION 3(a)(i), (ii)

- 3 Nowadays, fossil fuels are used to obtain heat. One of these fossil fuels is coal. Coal contains a large percentage by mass of carbon.

(a) (i) The complete combustion of coal is a redox reaction.

Write a balanced equation and show the oxidation number change for each element in the equation.

[3]

(ii) Identify the oxidising agent and the reducing agent.

oxidising agent

reducing agent

[1]

MARK SCHEME

3 (a) (i)	C + O ₂ → CO ₂ Oxidation number of carbon changes from 0 to +4 Oxidation number of oxygen changes from 0 to -2	1 1 1	ecf is considered
3 (a) (ii)	oxidising agent - oxygen reducing agent - carbon	1	Accept only both answers

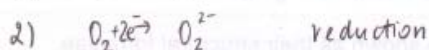
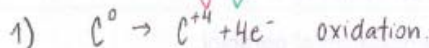
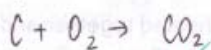
SAMPLE OF CANDIDATSE RESPONSE GRADED "A"

- 3 Nowadays, fossil fuels are used to obtain heat. One of these fossil fuels is coal. Coal contains a large percentage by mass of carbon.

For
Examiner's
Use

- (a) (i) The complete combustion of coal is a redox reaction.

Write a balanced equation and show the oxidation number change for each element in the equation.



[3]

3/3

- (ii) Identify the oxidising agent and the reducing agent.

oxidising agent oxygen

✓✓

reducing agent carbon

✓✓

[1]

1 1

EXAMINER'S COMMENTS

In question (a)(i), the candidate received three marks for the correct equation and the changes in oxidation states of carbon and oxygen;

In question (a)(ii), the candidate received one mark for correctly identifying the oxidising and the reducing agents.

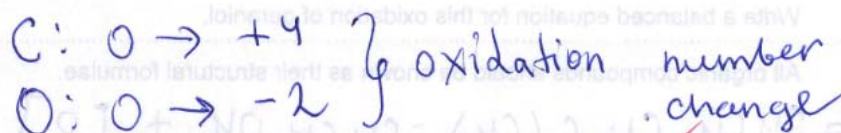
SAMPLE OF CANDIDATE RESPONSE GRADED "C"

- 3 Nowadays, fossil fuels are used to obtain heat. One of these fossil fuels is coal. Coal contains a large percentage by mass of carbon.

For
Examiner's
Use

- (a) (i) The complete combustion of coal is a redox reaction.

Write a balanced equation and show the oxidation number change for each element in the equation.



[3]

- (ii) Identify the oxidising agent and the reducing agent.

oxidising agent O_2

reducing agent O_2

[1]

EXAMINER'S COMMENTS

In question (a)(i), the candidate received three marks for the correct equation and the oxidation number of each element.

In question(a)(ii), the candidate lost a mark for incorrectly identifying the reducing agent.

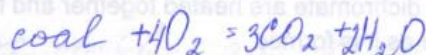
SAMPLE OF CANDIDATE RESPONSE GRADED "E"

- 3 Nowadays, fossil fuels are used to obtain heat. One of these fossil fuels is coal. Coal contains a large percentage by mass of carbon.

For
Examiner's
Use

- (a) (i) The complete combustion of coal is a redox reaction.

Write a balanced equation and show the oxidation number change for each element in the equation.



X X
X X
X X

[3]

- (ii) Identify the oxidising agent and the reducing agent.

oxidising agent O_2

reducing agent coal

[1]

EXAMINER'S COMMENTS

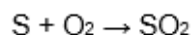
The candidate lost all marks in these questions.

In question (a)(i), the candidate did not write the correct equation, despite the condition stating that the main component of coal is carbon.

In question (a)(ii), the answer is incorrect because coal is a mixture, not an element.

QUESTION 3(b)

(b) A coal-fired power station uses a fuel that contains 0.5 % sulfur by mass.



Calculate the volume of sulfur dioxide, in dm^3 and measured at RTP, released by burning 20 tonnes of the fuel.

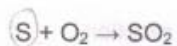
volume = dm^3 [4]

MARK SCHEME

3 (b)	$\text{S} + \text{O}_2 \rightarrow \text{SO}_2$ 1 : 1 : 1 20 tonnes = 20 000 000 g of fuel Mass of S = $20000000 \times 0.005 = 100000$ g $n(\text{S}) = 100000 / 32 = 3125$ moles 3125 moles sulfur produces 3125 moles of sulfur dioxide $V(\text{SO}_2) = 3125 \times 24 = 75000 \text{ dm}^3$	1 1 1 1	Accept if mass of sulfur given as 32.1 ecf is considered
-------	---	------------------	--

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

(b) A coal-fired power station uses a fuel that contains 0.5 % sulfur by mass.



Calculate the volume of sulfur dioxide, in dm^3 and measured at RTP, released by burning 20 tonnes of the fuel.

$$V = V_m \cdot n$$

$$V_m = 24 \text{ dm}^3 \text{ mol}^{-1}$$

$$n = \frac{m}{M} = \frac{20,000,000}{32} = 3125 \text{ mol}$$

$$M_r \text{ of S} \rightarrow 32.1$$

$$V = 24 \frac{\text{dm}^3}{\text{mol}} \times 3125 \text{ mol} = 75,000 \text{ dm}^3$$

$$20,000,000 \text{ g} - 100\%$$

$$\times - 0.5\%$$

$$x = 100,000 \text{ g}$$

$$\text{volume} = 75,000 \text{ dm}^3 [4]$$

EXAMINER'S COMMENTS

The candidate received full marks for the following:

Finding the mass of pure sulfur;

Finding the amount of sulfur;

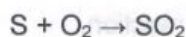
Finding the amount of sulfur dioxide;

Finding the volume of sulfur dioxide.

In addition, the student correctly converted the units.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

(b) A coal-fired power station uses a fuel that contains 0.5 % sulfur by mass.



Calculate the volume of sulfur dioxide, in dm^3 and measured at RTP, released by burning 20 tonnes of the fuel.

$$m = 20 \text{ tonnes} = 2 \cdot 10^7 \text{ g}$$

$$2 \cdot 10^7 - 100\%$$

$$\times - 0.5\% \Rightarrow x = 100,000 \text{ g} = 10^5 \text{ g} - \text{mass of sulfur}$$

$$n = \frac{m}{M} = \frac{10^5}{32} = 3125 \text{ mole}$$

$$n = \frac{V}{V_a} = \frac{V}{22.4} \Rightarrow 3125 \Rightarrow V = 70000 \text{ dm}^3$$

$$\text{volume} = 70000 \text{ dm}^3 [4]$$

EXAMINER'S COMMENTS

The candidate received three marks for the followings:

Finding the mass of pure sulfur;

Finding the amount of sulfur;

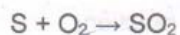
Finding the amount of sulfur dioxide.

The candidate lost one mark for finding volume of sulfur dioxide (using 22.4 at STP instead of 24 at RTP).

In addition, the student correctly converted units.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

- (b) A coal-fired power station uses a fuel that contains 0.5 % sulfur by mass.



Calculate the volume of sulfur dioxide, in dm^3 and measured at RTP, released by burning 20 tonnes of the fuel.

$$c = \frac{n}{V} \quad n = cV$$

$$m = 20000000 \text{ g} = 2 \cdot 10^7 \text{ g}$$

$$M(\text{SO}_2) = 32 + 32 = 64$$

$$1) n = \frac{m}{M} = \frac{2 \cdot 10^7}{64} = 3125 \cdot 10^2 \text{ mole}$$

$$2) V = \frac{n}{c} = \frac{3125 \cdot 10^2 \text{ mole}}{5 \cdot 10^{-3} \text{ mole dm}^{-3}} = 625 \cdot 10^5 \text{ dm}^3$$

$$\text{volume} = 625 \cdot 10^5 \text{ dm}^3 \quad [4]$$

EXAMINER'S COMMENTS

The candidate lost all marks for the followings reasons:

They did not find the mass of the pure component;

They incorrectly identified the main component of the fuel as SO_2 instead of S;

They incorrectly calculated the volume;

They did not use the equation.

The candidate correctly converted units, no extra marks were awarded for this.

QUESTION 3(c)

(c) Sulfur dioxide is toxic and is oxidised in the air to form sulfur trioxide.

Give **two** possible environmental problems caused by sulfur trioxide.

- 1 [1]
- 2 [1]

MARK SCHEME

3 (c)	Acid rain Lowering the pH of soil and lakes Killing trees Destroying of buildings	1 1	Any two of them. Accept any relevant answers.
-------	--	--------	---

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

(c) Sulfur dioxide is toxic and is oxidised in the air to form sulfur trioxide. SO_3

Give **two** possible environmental problems caused by sulfur trioxide.

1 Acid rains. SO_3 may react with water to form acid rains which cause deforestation. [1] 1 1

2 Eutrophication. SO_3 forms acid rain which results in increasing the acidity of lakes, rivers and death of habitat for fishes. SO_3 also may cause eutrophication. [1] 1 1

[Total 10] (10) (10)

Center for Pedagogical Measurements

EXAMINER'S COMMENTS

The candidate received all two marks for the topics 'acid rain' and 'increasing acidity of water'.

SAMPLE OF CANDIDATE RESPONSE GRADED “C”

(c) Sulfur dioxide is toxic and is oxidised in the air to form sulfur trioxide.

Give **two** possible environmental problems caused by sulfur trioxide.

- 1 acid rains ✓ [1]
- 2 toxic emissions on the atmosphere layer ✗ [1]

11
00

EXAMINER'S COMMENTS

The candidate received one mark for the topic of 'acid rain' but lost one mark for the second answer.

SAMPLE OF CANDIDATE RESPONSE GRADED “E”

(c) Sulfur dioxide is toxic and is oxidised in the air to form sulfur trioxide.

Give **two** possible environmental problems caused by sulfur trioxide.

- 1 green house effect, that may lead to the global warmth ✗ [1]
 - 2 respirational issues ✗ soil eruption ✗
 - 3 increase in solar radioactivity because the layer of gases, won't allow solar light to go back to the stratosphere ✗ [1]
- [Total 10]

00
00
①

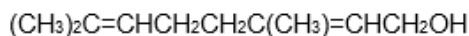
EXAMINER'S COMMENTS

The candidate lost all marks because they did not use the key words.

QUESTION 4(a) (i), (ii)

- 4 Geraniol and acidified potassium dichromate (IV) completely changes the odour of the compound.

The structural formula for geraniol is shown.



- (a) (i) Geraniol and acidified potassium dichromate are heated together and the product is distilled from the reaction mixture as it is formed.

Write a balanced equation for this oxidation of geraniol.

All organic compounds should be shown as their structural formulae.

[1]

- (ii) Describe the colour change observed during the reaction.

[1]

MARK SCHEME

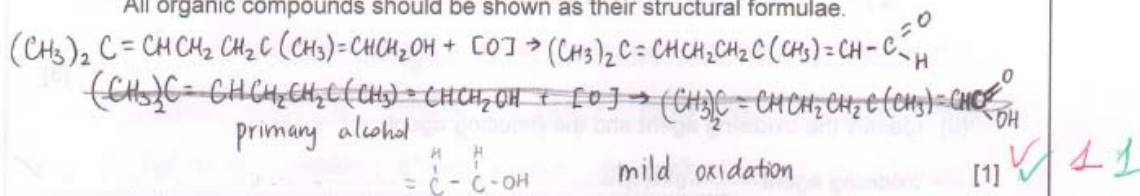
4 (a) (i)	$(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCH}_2\text{OH} + [\text{O}] \rightarrow (\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCHO} + \text{H}_2\text{O}$ $(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCH}_2\text{OH} + [\text{O}] \rightarrow (\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCHO}$	1	Accept only structural formulas of organic compounds
4 (a) (ii)	from orange to green /blue-green	1	Ignore colour change / only meaning of green colour without mentioning orange colour

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

- (a) (i) Geraniol and acidified potassium dichromate are heated together and the product is distilled from the reaction mixture as it is formed.

Write a balanced equation for this oxidation of geraniol.

All organic compounds should be shown as their structural formulae.



- (ii) Describe the colour change observed during the reaction.

from orange to green [1] ✓ 1 1

EXAMINER'S COMMENTS

**The candidate correctly drew the product;
The candidate correctly described the colour
change from orange to green.**

SAMPLE OF CANDIDATE RESPONSE GRADED “C”

4 Geraniol and acidified potassium dichromate (IV) completely changes the odour of the compound. The structural formula for geraniol is shown. $(CH_3)_2C=CHCH_2CH_2C(CH_3)=CHCH_2OH$ (a) (i) Geraniol and acidified potassium dichromate are heated together and the product is distilled from the reaction mixture as it is formed. Write a balanced equation for this oxidation of geraniol. All organic compounds should be shown as their structural formulae. $(CH_3)_2C=CHCH_2CH_2C(CH_3)=CHCH_2OH + K_2Cr_2O_7 \rightarrow (CH_3)_2C=CHCH_2CH_2C(CH_3)=CHCHO + H_2O$ – CH₂CHO (ii) Describe the colour change observed during the reaction. from orange to green	For Examiner's Use X [1] 0 ✓ ✓ [1] 1/1
---	--

EXAMINER'S COMMENTS

In question (a)(i), the candidate incorrectly defined the product of the reaction.

In question (a)(ii), the candidate correctly described the colour change.

SAMPLE OF CANDIDATE RESPONSE GRADED “E”

4	Geraniol and acidified potassium dichromate (IV) completely changes the odour of the compound.	For Examiner's Use
	The structural formula for geraniol is shown.	
	$(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)=\text{CHCH}_2\text{OH}$	
(a) (i)	Geraniol and acidified potassium dichromate are heated together and the product is distilled from the reaction mixture as it is formed.	
	Write a balanced equation for this oxidation of geraniol.	
	All organic compounds should be shown as their structural formulae.	
	$(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)=\text{CHCH}_2\text{OH} + \text{K}_2\text{Cr}_2\text{O}_7 \rightarrow$ $\rightarrow (\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)=\text{CHCH}_3 + \text{H}_2\text{O}$	[1] 00
(ii)	Describe the colour change observed during the reaction.	[1] X 00

EXAMINER'S COMMENTS

In question (a)(i), the candidate incorrectly defined the product of the reaction.
In question (a)(ii), the candidate incorrectly described the colour change.

QUESTION 4(b), (c) (i), (ii)

(b) It is possible to produce a carboxylic acid from geraniol.

State the condition needed for this reaction.

Write a balanced equation for this reaction.

All organic compounds should be shown as their structural formulae.

conditions

[1]

equation

[1]

(c) Geraniol reacts with a small amount of cold acidified dilute potassium manganate (VII).

(i) State which functional group will react in this reaction.

..... [1]

(ii) State how many hydroxyl groups are present in the organic product of this reaction.

..... [1]

MARK SCHEME

4 (b)	reflux $(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCH}_2\text{OH} + 2[\text{O}] \rightarrow (\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCOOH} + \text{H}_2\text{O}$ Or $(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCHO} + [\text{O}] \rightarrow (\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)\text{CHCOOH}$	1 1	Accept only structural formulas of organic compounds
4 (c) (i)	Carbon-carbon double bond/ alkene	1	
4(c) (ii)	5/five	1	

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

(b) It is possible to produce a carboxylic acid from geraniol.

State the condition needed for this reaction.

Write a balanced equation for this reaction.

All organic compounds should be shown as their structural formulae.

conditions heat under reflux ✓, acidified potassium dichromate

equation

$$\text{CH}_3-\underset{\text{CH}_3}{\text{C}}=\text{CH}-\text{CH}_2-\text{CH}_2-\underset{\text{CH}_3}{\text{C}}=\text{CH}-\text{CH}_2-\text{OH} + 2[\text{O}] \xrightarrow[\text{reflux}]{\text{KMnO}_4/\text{H}^+} \text{CH}_3-\underset{\text{CH}_3}{\text{C}}=\text{CH}-\text{CH}_2-\text{CH}_2-\underset{\text{CH}_3}{\text{C}}=\text{CH}-\overset{\text{O}}{\underset{\text{OH}}{\text{C}}} + \text{H}_2\text{O} \quad \checkmark$$

(c) Geraniol reacts with a small amount of cold acidified dilute potassium manganate (VII).

(i) State which functional group will react in this reaction.

C=C alkene double bond ✓ [1]

(ii) State how many hydroxyl groups are present in the organic product of this reaction.

1 ✓ [1]

EXAMINER'S COMMENTS

The candidate received two marks for (b). They correctly mentioned "reflux" as a condition and identified the product of the reaction.

The candidate received two marks for (c). They stated the functional group in (c)(i) and the number of hydroxyl groups involved in the reaction.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

(b) It is possible to produce a carboxylic acid from geraniol.

State the condition needed for this reaction.

Write a balanced equation for this reaction.

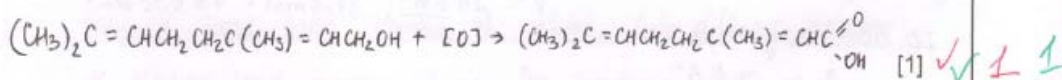
All organic compounds should be shown as their structural formulae.

conditions

H_2SO_4 as catalyst $\text{H}^+/\text{Cr}_2\text{O}_7^{2-}$
heat and reflux ✓✓
equation

[1]

1 1



(c) Geraniol reacts with a small amount of cold acidified dilute potassium manganate (VII).

(i) State which functional group will react in this reaction. KMnO_4

OH group

[1]

X X 0 0

(ii) State how many hydroxyl groups are present in the organic product of this reaction.

0

[1]

X X 0 0

EXAMINER'S COMMENTS

The candidate received marks for (b) as they correctly wrote 'reflux' as a condition and identified the product of the reaction.

However, the candidate was unable to state the functional group in (c)(i) and the number of hydroxyl groups involved in the reaction.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

- (b) It is possible to produce a carboxylic acid from geraniol.

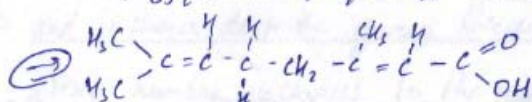
State the condition needed for this reaction.

Write a balanced equation for this reaction.

All organic compounds should be shown as their structural formulae.

conditions ^{high} temperature, catalyst (KMnO_4 , V_2O_5 , Fe)

equation $(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)=\text{CHCH}_2\text{OH} + \text{H}_2 \rightarrow$



- (c) Geraniol reacts with a small amount of cold acidified dilute potassium manganate (VII).

- (i) State which functional group will react in this reaction.

hydroxyl group

- (ii) State how many hydroxyl groups are present in the organic product of this reaction.

1

EXAMINER'S COMMENTS

The candidate did not receive any marks.

In question (b), the candidate mistakenly wrote "temperature" instead of "reflux", which has a different meaning. Additionally, the equation provided was incorrect.

In question (c), the candidate did not state the functional group and the number of hydroxyl groups in the product.

QUESTION 4(d)

(d) State what type of stereoisomerism is in the product

.....

Explain your answer.

.....

.....

.....

[2]

MARK SCHEME

4(d)	optical isomerism Molecule has chiral atoms/assymmetric carbon atoms/four different groups or atoms $(\text{CH}_3)_2\text{C}(\text{OH})\text{C}^*\text{H}(\text{OH})\text{CH}_2\text{CH}_2\text{C}^*(\text{CH}_3)(\text{OH})\text{C}^*\text{H}(\text{OH})\text{CH}_2\text{OH}$	1 1	
------	--	-------------------	--

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

7

(d) State what type of stereoisomerism is in the product

optical isomerism

Explain your answer.

because there will be chiral carbon atoms attached
to 4 different functional groups.

For
Examiner's
Use

[2]

22

EXAMINER'S COMMENTS

The candidate stated the type of isomerism correctly and explained it using the concept of a chiral carbon.

SAMPLE OF CANDIDATE RESSPONSE GRADED "C"

(d) State what type of stereoisomerism is in the product

optical

For
Examiner's
Use

Explain your answer.

because it has mirror image of the enantiomers that
are 2 opti; only 2 optical isomers.

[2]

1

EXAMINER'S COMMENTS

The candidate correctly stated the type of isomerism but could not explain why they chose it.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

(d) State what type of stereoisomerism is in the product

geometrical

For
Examiner's
Use

Explain your answer.

- 1) different group on right-hand side ^{and} left-hand side of double bond.
- 2) restricted rotation of C=C bond.

[2]

0.

EXAMINER'S COMMENTS

The candidate could not correctly identify the type of isomerism correctly and explain it.

QUESTION 5 (a), (b)

5 Benzene reacts with ethanoyl chloride in a Friedel-Crafts acylation.

(a) Complete the equation for the reaction between C_6H_6 and CH_3COCl .

$C_6H_6 + CH_3COCl \rightarrow$ [1]

Write the names of the products of the reaction.

..... [1]

(b) The reaction between benzene and ethanoyl chloride uses an aluminium chloride catalyst.

Name the type of reaction and state one **other** reaction condition.

reaction type [1]

condition [1]

MARK SCHEME

5 (a)	$C_6H_6 + CH_3COCl \rightarrow C_6H_5COCH_3 + HCl$	1	Accept only in structural or displayed formulae of products. Do NOT accept molecular formula of the main product.
	phenylethanone and hydrogen chloride	1	Accept if only phenylethanone is given.
5 (b)	electrophilic substitution / acylation	1	
	Temperature – 60 °C/heating under reflux	1	Accept range 40-80°C

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

5 Benzene reacts with <u>ethanoyl chloride</u> in a Friedel-Crafts acylation. (a) Complete the equation for the reaction between C_6H_6 and CH_3COCl. $C_6H_6 + CH_3COCl \rightarrow$ <u>$C_6H_5COCH_3 + HCl$</u> ✓ [1] Write the names of the products of the reaction. <u>hydrochloric acid, phenyl ketone (phenyl ethanone)</u> ✓ [1] (b) The reaction between benzene and ethanoyl chloride uses an aluminium chloride catalyst. Name the type of reaction and state one other reaction condition. reaction type <u>electrophilic substitution</u> ✓ [1] condition <u>heat reflux</u> ✓ [1]	For Examiner's Use / / / /
---	---

EXAMINER'S COMMENTS

In question (a), the candidate correctly wrote the equation and the names of the products, earning them two marks.

In question (b), the candidate accurately identified the reaction type and provided the correct conditions.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

5 Benzene reacts with ethanoyl chloride in a Friedel-Crafts acylation. (a) Complete the equation for the reaction between C_6H_6 and CH_3COCl. $C_6H_6 + CH_3COCl \rightarrow$ <u>ethanoic acid $CH_3COC_6H_5 + HCl$</u> [1] ✓ 1 1 Write the names of the products of the reaction. <u>hydrochloric acid and benzylethanoic acid</u> [1] ✗ 0 0 (b) The reaction between benzene and ethanoyl chloride uses an aluminium chloride catalyst. Name the type of reaction and state one other reaction condition. reaction type <u>electrophillic addition</u> [1] ✗ 0 0 condition <u>reflux and heat</u> [1] 1 1	For Examiner's Use
---	-----------------------------------

Handwritten note on the left: $CH_3-C(=O)-C_6H_5$

EXAMINER'S COMMENTS

In (a) question, the candidate received one mark for the correct equation but lost one mark for the incorrect IUPAC naming of it.

In (b) question, the candidate lost one mark due to the incorrect reaction type, but the candidate earned one mark for the correct condition.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

5	Benzene reacts with ethanoyl chloride in a Friedel-Crafts acylation.	For Examiner's Use
(a)	Complete the equation for the reaction between C_6H_6 and CH_3COCl .	
	$C_6H_6 + CH_3COCl \rightarrow$ <u>$CH_3CH_2CH_2CH_2CH_2CH_3$ $CH_3C_6H_5 + CO + Cl$</u> X [1]	0
	Write the names of the products of the reaction.	
	<u>chloromethylbenzene ethanoyl methyl ben zene</u> X [1] <u>methyl ben zene, ethano</u>	0
(b)	The reaction between benzene and ethanoyl chloride uses an aluminium chloride catalyst.	
	Name the type of reaction and state one other reaction condition.	
	reaction type <u>addition</u> X [1]	0
	condition <u>optimal temperature, catalyst ($AlCl_3$)</u> X [1]	0

EXAMINER'S COMMENTS

In question (a), the candidate received no marks for the incorrect equation and the incorrect IUPAC naming of the products.

In question (b), the candidate lost all marks due to the incorrect reaction type and the incorrect conditions for Friedel-Crafts acylation.

QUESTION 5 (c), (d)

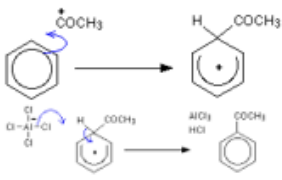
- (c) Complete the equation for the reaction between ethanoyl chloride and aluminium chloride:



- (d) Draw the mechanism for the reaction in 5(a).

[4]

MARK SCHEME

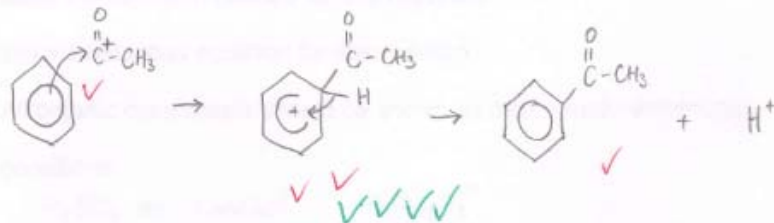
5 (c)	$\text{CH}_3\text{COCl} + \text{AlCl}_3 \longrightarrow \text{CH}_3\text{CO}^+ + \text{AlCl}_4^-$	1	Do NOT accept if charges are not shown.
5 (d)	 A curly arrow from benzene ring to the electrophile For partially breaking of the delocalized electron ring in benzene molecule A curly arrow from AlCl_4^- to the appropriate hydrogen of the benzene ring OR A curly arrow from C-H bond of the ring to partially broken delocalized electron ring Correct products	1 1 1 1	

SAMPLE OF CANDIDATE RESPONSE GRADED "A"

- (c) Complete the equation for the reaction between ethanoyl chloride and aluminium chloride:



- (d) Draw the mechanism for the reaction in 5(a).



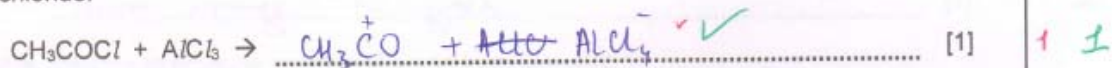
[4] 4 4

EXAMINER'S COMMENTS

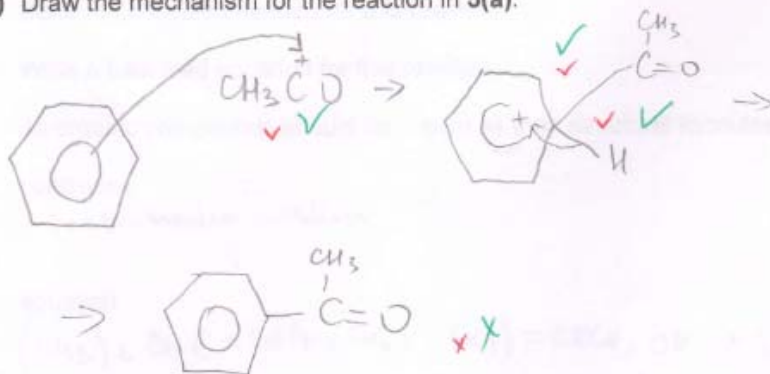
The candidate received one mark for correctly balancing the equation between ethanoyl chloride and the catalyst;
The candidate drew all steps in mechanisms correctly.

SAMPLE OF CANDIDATE RESPONSE GRADED "C"

- (c) Complete the equation for the reaction between ethanoyl chloride and aluminium chloride:



- (d) Draw the mechanism for the reaction in 5(a).



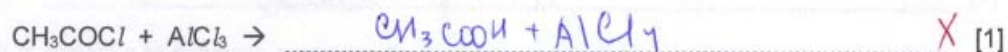
[4] 3 3

EXAMINER'S COMMENTS

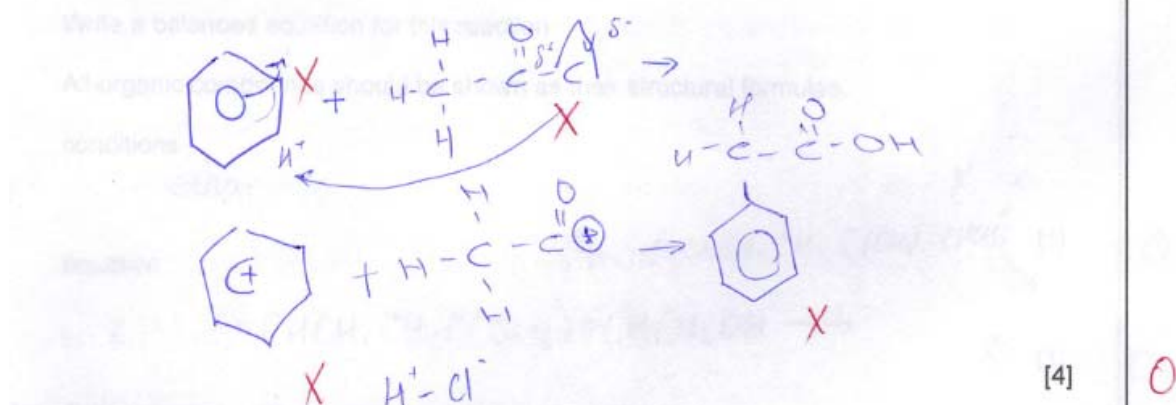
The candidate received one mark for correctly balancing the equation between ethanoyl chloride and the catalyst;
The candidate drew all steps in the mechanism correctly, but the hydrogen ion is absent in the product.

SAMPLE OF CANDIDATE RESPONSE GRADED "E"

- (c) Complete the equation for the reaction between ethanoyl chloride and aluminium chloride:



- (d) Draw the mechanism for the reaction in 5(a).



EXAMINER'S COMMENTS

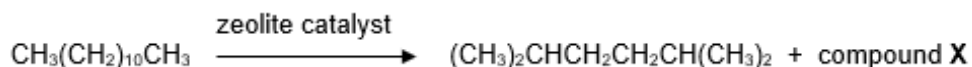
The candidate received no marks for this question. The equation is entirely incorrect as a mechanism.

QUESTION 6(a) (i), (ii), (iii)

6 Dodecane, $C_{12}H_{26}$, is found in the kerosene fraction of crude oil.

E

The catalytic cracking of a dodecane molecule gives one branched alkane and one straight chain alkene molecule.



(a) Catalytic cracking involves bond breaking known as heterolytic fission. Heterolytic fission produces positive and negative ions.

(i) Define the term catalytic cracking.

.....
[1]

(ii) Write the structural formula of the compound X.

[1]

(iii) Write an equation to show an example of heterolytic fission of a dodecane molecule.

Include electrons in the structures of the products of the fission.

[2]

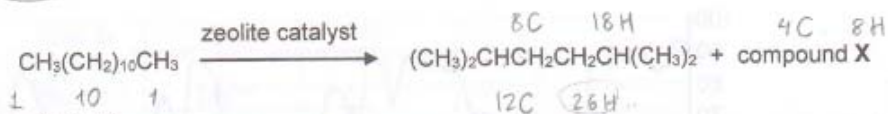
MARK SCHEME

6 (a) (i)	using catalyst to break long chain hydrocarbon to smaller hydrocarbons.	1	Owtte										
6 (a) (ii)	CH ₃ CH=CHCH ₃ / CH ₃ CH ₂ CH=CH ₂	1	Accept only structural formula or displayed formula										
6 (a) (iii)	correct ion with positive charge correct ion with negative charge and a pair of outer shell electrons Examples <table border="1"> <thead> <tr> <th>Negative charged ion (with electron pair)</th> <th>Positive charged ion</th> </tr> </thead> <tbody> <tr> <td>CH₃CH₂CH₂CH₂CH₂CH₂C H₂CH₂CH₂CH₂CH₂:⁻</td> <td>CH₃⁺</td> </tr> <tr> <td>CH₃CH₂CH₂CH₂CH₂CH₂C H₂CH₂CH₂CH₂:⁻</td> <td>CH₃CH₂⁺</td> </tr> <tr> <td>CH₃CH₂CH₂CH₂CH₂CH₂C H₂CH₂CH₂:⁻</td> <td>CH₃CH₂CH₂⁺</td> </tr> <tr> <td>CH₃CH₂CH₂CH₂CH₂CH₂C H₂CH₂:⁻</td> <td>CH₃CH₂CH₂CH₂⁺</td> </tr> </tbody> </table>	Negative charged ion (with electron pair)	Positive charged ion	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ C H ₂ CH ₂ CH ₂ CH ₂ CH ₂ : ⁻	CH ₃ ⁺	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ C H ₂ CH ₂ CH ₂ CH ₂ : ⁻	CH ₃ CH ₂ ⁺	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ C H ₂ CH ₂ CH ₂ : ⁻	CH ₃ CH ₂ CH ₂ ⁺	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ C H ₂ CH ₂ : ⁻	CH ₃ CH ₂ CH ₂ CH ₂ ⁺	1 1	If calculate carbons / hydrogens of two ions should be C₁₂H₂₆ Do not accept another molecule than dodecane for giving full 2 marks
Negative charged ion (with electron pair)	Positive charged ion												
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ C H ₂ CH ₂ CH ₂ CH ₂ CH ₂ : ⁻	CH ₃ ⁺												
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ C H ₂ CH ₂ CH ₂ CH ₂ : ⁻	CH ₃ CH ₂ ⁺												
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ C H ₂ CH ₂ CH ₂ : ⁻	CH ₃ CH ₂ CH ₂ ⁺												
CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₂ C H ₂ CH ₂ : ⁻	CH ₃ CH ₂ CH ₂ CH ₂ ⁺												

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

- 6 Dodecane, $C_{12}H_{26}$, is found in the kerosene fraction of crude oil.

The catalytic cracking of a dodecane molecule gives one branched alkane and one straight chain alkene molecule.

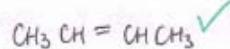
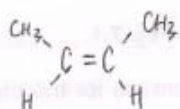
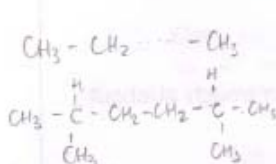


- (a) Catalytic cracking involves bond breaking known as heterolytic fission. Heterolytic fission produces positive and negative ions.

- (i) Define the term catalytic cracking.

Catalytic cracking is the separating long hydrocarbon chain into smaller chains by using of a catalyst. [1] ✓

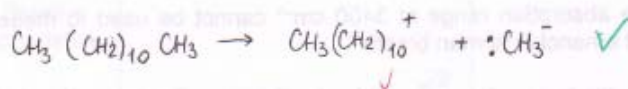
- (ii) Write the structural formula of the compound X.



[1] ✓

- (iii) Write an equation to show an example of heterolytic fission of a dodecane molecule.

Include electrons in the structures of the products of the fission.



[2] ✓

EXAMINER'S COMMENTS

The candidate received all four marks.

In question (a)(i), the student correctly wrote the description using the key words.

In question (a)(ii), the student provided the structural formula of compound X.

In question (a)(iii), the student correctly illustrated the two products of the heterolytic fission, including the representation of the lone pair and charges.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

(a) Catalytic cracking involves bond breaking known as heterolytic fission. Heterolytic fission produces positive and negative ions.

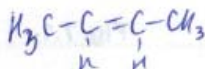
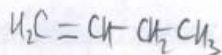
(i) Define the term catalytic cracking.

breaking down the long chains of ~~car~~ hydrocarbons (alkanes)
by using catalyst

[1] ✓

1

(ii) Write the structural formula of the compound X.

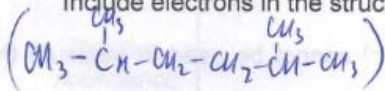


[1] ✓

1

(iii) Write an equation to show an example of heterolytic fission of a dodecane molecule.

Include electrons in the structures of the products of the fission.



[2] X

0

EXAMINER'S COMMENTS

The candidate received two out of four marks.

In question (a)(i), the student correctly wrote the description using the key words.

In question (a)(ii), the student provided the structural formula of compound X.

In question (a)(iii), the student did not illustrate the two products of the heterolytic fission, including the representation of the lone pair and charges.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

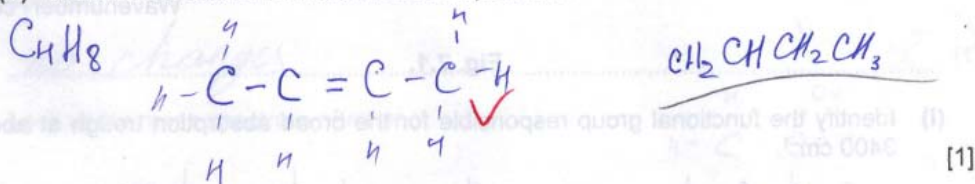
(a) Catalytic cracking involves bond breaking known as heterolytic fission. Heterolytic fission produces positive and negative ions.

(i) Define the term catalytic cracking.

Is the process of alkene C=C bond breaking using a catalyst

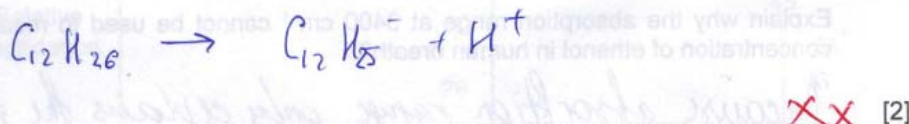
X [1]

(ii) Write the structural formula of the compound X.



(iii) Write an equation to show an example of heterolytic fission of a dodecane molecule.

Include electrons in the structures of the products of the fission.



EXAMINER'S COMMENTS

The candidate received one out of four marks.

In question (a)(i), the student incorrectly wrote the description without using the key words.

In question (a)(ii), the student provided the structural formula of compound X.

In question (a)(iii), the student illustrated the two products of the heterolytic fission, including charges. However, the CH bond is too strong to split before the CC bond, which means that the equation is not correct.

QUESTION 6(b), (c)

(b) Explain the industrial importance of cracking.

.....

.....

.....

..... [2]

(c) The main difference between *thermal* cracking and *catalytic* cracking is the way in which covalent bonds in the hydrocarbon chains are broken.

Explain the meaning of *homolytic fission*.

.....

..... [1]

MARK SCHEME

6 (b)	Main Answer: In our modern world low fractions like petrol in a high demand than supply from rectification of crude oil. Due to that reason a catalytic cracking is used to obtain low hydrocarbon fractions from heavier fractions of crude oil. Credit only 1 mark for the following one or several ideas: for making alkenes that are monomers for making polymers / as a source of hydrogen / obtaining precursors for chemicals etc.	1 1	owtte
6 (c)	unpaired electron/the splitting of the covalent bond produces two free radicals/ equal splitting	1	Accept appropriate diagram illustration

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

(b) Explain the industrial importance of cracking.

Cracking used in fractional distillation of crude oil. Long hydrocarbon chains may be split into smaller hydrocarbon chains and distilled due to the molecular mass, that helps to have a product as petrol, kerosene and etc. Different hydrocarbon chain length results in differences of physical state, chemical properties that increase the variety of products. [2]

(c) The main difference between thermal cracking and catalytic cracking is the way in which covalent bonds in the hydrocarbon chains are broken.

Explain the meaning of homolytic fission.

Homolytic fission is equal division of shared electron pair to each atom. results in forming of radical. Example: $A-B \rightarrow A\cdot + B\cdot$ [1]

EXAMINER'S COMMENTS

The candidate received all three marks.

In question (b), the candidate presented two ideas: the small area of uses for hydrocarbons with long chains and the importance of small hydrocarbons.

In question (c), the candidate explained the meaning of homolytic fission using the key words and provided the correct equation.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

(b) Explain the industrial importance of cracking.

Cracking is the most popular way of producing hydrocarbons like alkanes and alkenes. It's importance is that often we can't use long chained hydrocarbons produced in fractional distillation, so we need to break them into smaller molecules that can be used in vehicles, lighting, etc. [2]

(c) The main difference between thermal cracking and catalytic cracking is the way in which covalent bonds in the hydrocarbon chains are broken.

Explain the meaning of homolytic fission.

Bond breaking where each product has one electron from shared electron pair left, meaning that covalent bond is broken equally and 2 radicals are formed. [1]

EXAMINER'S COMMENTS

The candidate received two out of three marks.

In question (b), the candidate only presented one idea about the importance of small hydrocarbons.

In question (c), the candidate explained the meaning of homolytic fission using the key words.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

(b) Explain the industrial importance of cracking.

Production of alkanes using in pharmaceuticals,
also production of crude oils using as energy
oils in car

[2]

(c) The main difference between *thermal* cracking and *catalytic* cracking is the way in which covalent bonds in the hydrocarbon chains are broken.

Explain the meaning of *homolytic fission*.

Producing of only negative ions

[1]

EXAMINER'S COMMENTS

The candidate received only one mark.

The candidate provided one example of using the product of cracking but did not include the main ideas in the answer. In question (c), the candidate did not explain the meaning of homolytic fission.

QUESTION 7(a)(i), (ii)

- 7 A range of different analytical techniques such as NMR, IR spectroscopy and mass spectrometry are used to detect and identify substances.

(a) Look at the infrared spectrum of ethanol in Fig. 7.1.

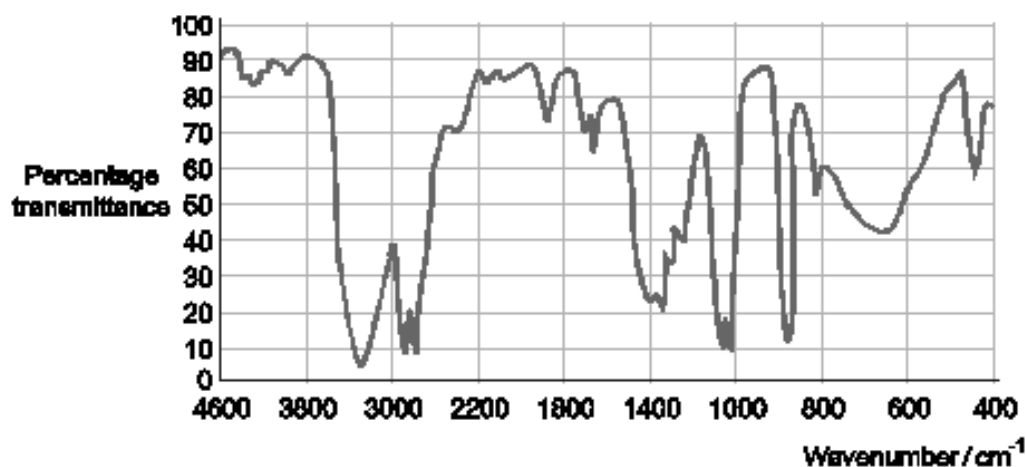


Fig. 7.1.

- (i) Identify the functional group responsible for the broad absorption trough at about 3400 cm^{-1} .

..... [1]

- (ii) Human breath contains water vapor.

Explain why the absorption range at 3400 cm^{-1} cannot be used to measure the concentration of ethanol in human breath?

..... [1]

MARK SCHEME

7 a (i)	–OH in alcohols/ hydroxyl group in alcohols	1	
7 a (ii)	Water also has –OH group	1	

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

- (i) Identify the functional group responsible for the broad absorption trough at about 3400 cm^{-1} .

-OH group

[1]

- (ii) Human breath contains water vapor.

Explain why the absorption range at 3400 cm^{-1} cannot be used to measure the concentration of ethanol in human breath?

because water also contains O-H bond.

[1]

EXAMINER'S COMMENTS

The candidate received all two marks.

The candidate correctly identified the hydroxyl group based on the IF data and clearly showed that water also contains an OH group.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

- 7 A range of different analytical techniques such as NMR, IR spectroscopy and mass spectrometry are used to detect and identify substances.

For
Examiner's
Use

(a) Look at the infrared spectrum of ethanol in Fig. 7.1.

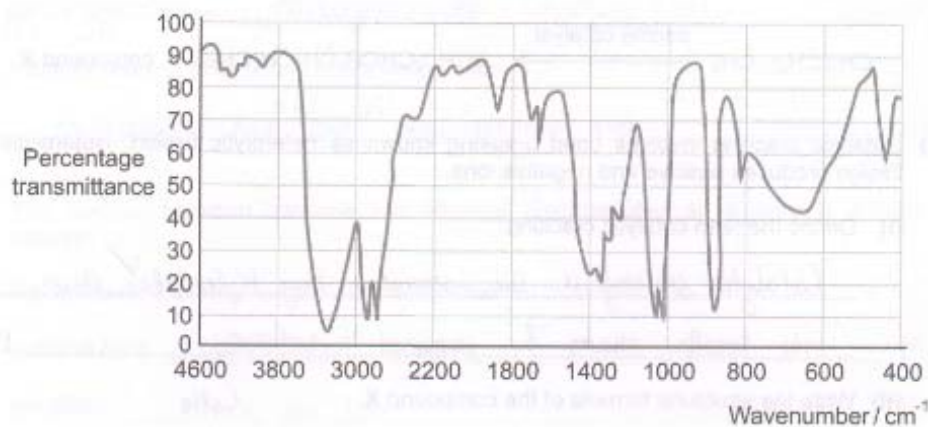


Fig. 7.1.

- (i) Identify the functional group responsible for the broad absorption trough at about 3400 cm⁻¹.

hydroxyl group (-OH)

[1]

✓ 1 1

- (ii) Human breath contains water vapor.

Explain why the absorption range at 3400 cm⁻¹ cannot be used to measure the concentration of ethanol in human breath?

Infrared spectrum is used only to determine the type of bonds (not the concentration), each specific bond vibrates and it can be detected by absorbing infrared radiation.

[1]

X 0 0

EXAMINER'S COMMENTS

The candidate received only one mark.

The candidate correctly identified the hydroxyl group based on the IR data but could not understand that water also contains an OH group.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

- (i) Identify the functional group responsible for the broad absorption trough at about 3400 cm^{-1} .

ester

X [1]

- (ii) Human breath contains water vapor.

Explain why the absorption range at 3400 cm^{-1} cannot be used to measure the concentration of ethanol in human breath?

Because it has very low percentage of transmittance, so the results could be hard to identify with a lot of errors

X [1]

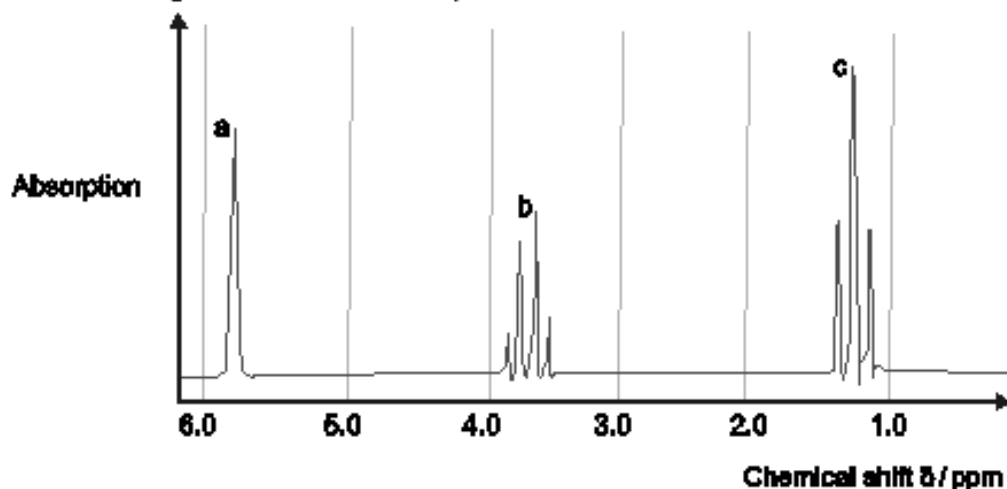
EXAMINER'S COMMENTS

The candidate received no mark.

The candidate incorrectly identified the functional group based on the IF data and could not understand that water also contains an OH group.

QUESTION 7(b)(i), (ii) (c)

(b) Look at the high-resolution ^1H NMR spectrum of ethanol.



(i) Define the parts of the ethanol molecule with hydrogen atoms corresponding to the peaks:

Peak a

Peak b

Peak c [2]

(ii) Explain the meaning of splitting in ^1H NMR spectroscopy using the $n+1$ rule.

..... [1]

(c) Heavy water, D_2O , is used to identify functional groups with labile protons.

Describe the change in the high resolution ^1H NMR spectrum of ethanol when it has D_2O added.

..... [1]

MARK SCHEME

7 (b) (i)	Peak a is $-\text{OH}$ Peak b $-\text{CH}_2-$ Peak c $-\text{CH}_3$	1 1 1 Max [2]	M2 – for three correct answers M1 – for two correct answers
7 (b) (ii)	Peaks always split into the number of hydrogens on the neighbouring/adjacent carbon(s), plus one.	1	
7 (c)	The peak with respect to $-\text{OH}$ group disappears / would be two peaks only	1	

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

(i) Define the parts of the ethanol molecule with hydrogen atoms corresponding to the peaks:

Peak a OH ✓

Peak b CH₂ ✓

Peak c CH₃ [2]

(ii) Explain the meaning of splitting in ¹H NMR spectroscopy using the n+1 rule.

The splitting shows the amount of neighboring hydrogens, but an additional peak is shown on spectrum ✓ [1]

(c) Heavy water, D₂O, is used to identify functional groups with labile protons.

Describe the change in the high resolution ¹H NMR spectrum of ethanol when it has D₂O added.

peak a disappears ✓ [1]

EXAMINER'S COMMENTS

The candidate received full marks.

In question (b)(i), the candidate successfully demonstrated all three parts.

In question (b)(ii), the candidate explained the meaning of "n+1" using the provided key word.

In question (c)(ii), the candidate accurately described the effect on the peak when deuterium water was added.

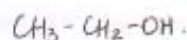
SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

- (i) Define the parts of the ethanol molecule with hydrogen atoms corresponding to the peaks:

Peak a - OH ✓ ✓

Peak b - CH₂ - ✓ ✓

Peak c CH₃ - ✓ ✓



For
Examiner's
Use

[2]

2/2

- (ii) Explain the meaning of splitting in ¹H NMR spectroscopy using the n+1 rule.

Each carbon atom shows the hydrogen atoms of adjacent carbon atoms. The substance is located in CDCl₃ which shows one peak. [1]

1 1

- (c) Heavy water, D₂O, is used to identify functional groups with labile protons.

Describe the change in the high resolution ¹H NMR spectrum of ethanol when it has D₂O added.

~~There~~ no change because ¹H works only with H atoms (H behaves as a magnet). [1]

X 0 0

EXAMINER'S COMMENTS

The candidate received three out of four marks.

In question (b)(i), the candidate demonstrated all three parts successfully.

In question (b)(ii), the candidate explained the meaning of "n+1" using the provided key word.

In question (c)(ii), the candidate did not describe the changes that occurred when deuterium water was added.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

(i) Define the parts of the ethanol molecule with hydrogen atoms corresponding to the peaks:		For Examiner's Use
Peak a <u>C-O</u> X	$ \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{O}-\text{CH}_3 \\ \\ \text{H} \end{array} $	
Peak b <u>CH₃</u> X		
Peak c <u>-O-CH</u> X		[2] 0
(ii) Explain the meaning of splitting in ¹ H NMR spectroscopy using the n+1 rule.		
<u>the resolution with the different splittings, so the n+1 will show ending result</u> X		[1] 0
(c) Heavy water, D ₂ O, is used to identify functional groups with labile protons.		
Describe the change in the high resolution ¹ H NMR spectrum of ethanol when it has D ₂ O added.		
<u>there will be more contrast and more difference between peaks</u> X		[1] 0

EXAMINER'S COMMENTS

The candidate received no marks.

In question (b)(i), the candidate incorrectly wrote parts of the molecule.

In question (b)(ii), the candidate did not explain the meaning of "n+1".

In question (c)(ii), the candidate did not describe the changes that occurred when deuterium water was added.

QUESTION 7(d)

(d) Look at the mass spectrum of ethanol.

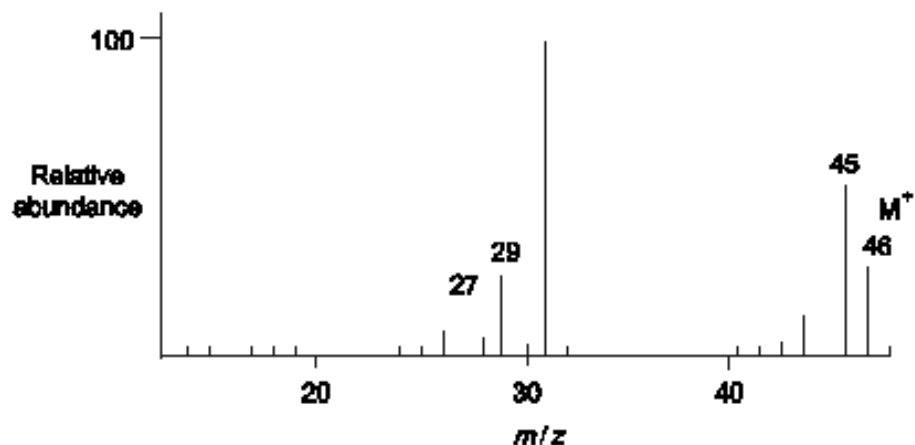


Fig. 7.3.

(i) Write the formulae of the particles with the m/z value shown.

$m/z = 29$, corresponding particle

$m/z = 45$, corresponding particle [2]

(ii) Write the meaning of the M^+ peak in mass spectrometry.

..... [1]

MARK SCHEME

7 (d) (i)	$m/z=29$, corresponding ion is CH_3CH_2^+ / C_2H_5^+ $m/z=45$, corresponding ion is $\text{CH}_3\text{CH}_2\text{O}^+$ / $\text{C}_2\text{H}_5\text{O}^+$	1 1	Do NOT accept without charge
7 (d) (ii)	Relative molecular mass of the compound/ molecular ion peak	1	The answer is accepted if the word 'relative' is missed.

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

(i) Write the formulae of the particles with the m/z value shown.		
$m/z = 29$, corresponding particle	CH_3CH_2^+ ✓ ✓	
$m/z = 45$, corresponding particle	$\text{CH}_3\text{CH}_2\text{O}^+$ ✓ ✓ [2]	2 2
(ii) Write the meaning of the M^+ peak in mass spectrometry.		
	the whole ethanol molecule's peak X ✓ [1]	0 1

EXAMINER'S COMMENTS

The candidate received full marks.

In question (d)(i), the candidate correctly defined particles from mass spectra, including their charges.

In question (d)(ii), the candidate used the key phrase 'molecular peak'.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

(i) Write the formulae of the particles with the m/z value shown.		
$m/z = 29$, corresponding particle	CH_3CH_2^+ ✓ ✓	
$m/z = 45$, corresponding particle	$\text{CH}_3\text{CH}_2\text{O}^+$ ✓ ✓ [2]	2 2
(ii) Write the meaning of the M^+ peak in mass spectrometry.		
	M^+ shows the presence of a C isotope. XX [1] X	0 0

EXAMINER'S COMMENTS

The candidate received three marks out of four.

The candidate correctly identified the two particles from mass spectra data, including charges, which are important for it. However, the candidate couldn't describe what M^+ in mass spectra means.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

(i) Write the formulae of the particles with the m/z value shown.		
$m/z = 29$, corresponding particle	CH_3 X	
$m/z = 45$, corresponding particle	CH_3OCH X [2]	0
(ii) Write the meaning of the M^+ peak in mass spectrometry.		
	the highest point of the energetical results between the bands of a compound. X [1]	0
		[Total 9] 0

EXAMINER'S COMMENTS

The candidate received no marks.

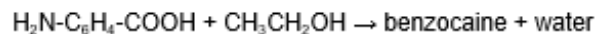
The candidate did not identify the particles correctly, including charges.

Moreover, the definition of the M^+ peak is incorrect.

QUESTION 8(a) (b) (c) (i), (ii)

- 8 Benzocaine is a local anesthetic, which means that it makes only small areas of the body insensitive to pain.

The preparation of benzocaine by the esterification of 4-aminobenzoic acid with ethanol is shown.



- (a) State conditions for the reaction above.

..... [2]

- (b) Write the structural formula for benzocaine.

[1]

- (c) (i) Name all the functional groups in benzocaine.

..... [2]

- (ii) Give the systematic name of benzocaine.

..... [1]

MARK SCHEME

8 (a)	conc. sulphuric acid	1	Accept, only concentrated sulfuric acid
	reflux/heat	1	
8 (b)	$\text{H}_2\text{N}-\text{C}_6\text{H}_4-\text{COOCH}_2\text{CH}_3$	1	Accept only structural formula or displayed formula
8 (c) (i)	ester, amine, benzene ring	1 1 1 [Max. 2]	M2 – for three correct answers
			M1 – for two correct answers
8 (c) (ii)	ethyl-4-aminobenzoate	1	do NOT accept 4-amino-ethylbenzoate

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

- 8 Benzocaine is a local anesthetic, which means that it makes only small areas of the body insensitive to pain.

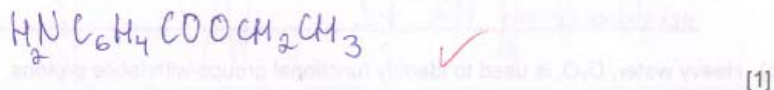
The preparation of benzocaine by the esterification of 4-aminobenzoic acid with ethanol is shown.



- (a) State conditions for the reaction above.

concentrated H_2SO_4 , reflux, temperature [2]

- (b) Write the structural formula for benzocaine.



- (c) (i) Name all the functional groups in benzocaine.

amine, benzene, ester bond ← carboxyl ~~ether~~ [2]

- (ii) Give the systematic name of benzocaine.

ethyl-4-aminobenzoate [1]

For
Examiner's
Use

2

1

2

1

EXAMINER'S COMMENTS

This candidate received full marks.

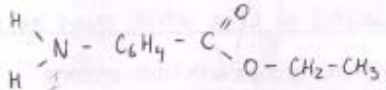
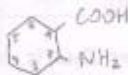
In question (a), the candidate correctly wrote the conditions.

In question (b), the candidate showed the correct formula for benzocaine.

In question (c)(i), the candidate named all three functional groups correctly.

In question (c)(ii), the systematic name of benzocaine was written correctly.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

8	Benzocaine is a local anesthetic, which means that it makes only small areas of the body insensitive to pain.	For Examiner's Use
	The preparation of benzocaine by the esterification of 4-aminobenzoic acid with ethanol is shown.	
	$\text{H}_2\text{N}-\text{C}_6\text{H}_4-\text{COOH} + \text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{benzocaine} + \text{water}$	
(a)	State conditions for the reaction above.	
	H_2SO_4 as catalyst, heat and reflux [2]	1 1
(b)	Write the structural formula for benzocaine.	
	  [1]	1 1
(c) (i)	Name all the functional groups in benzocaine.	
	1) amino group ✓✓ 3) carbonyl group (C=O) ✗✗ 2) alkyl alkyne group ester 4) alkyl group ✗✗ [2]	0 0
(ii)	Give the systematic name of benzocaine.	
	2-amino benzoic acid ✗ [1]	0 0

EXAMINER'S COMMENTS

The candidate received only two marks out of six.

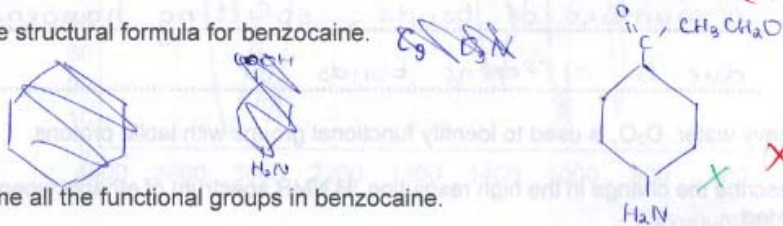
In question (a), the candidate correctly wrote the reflux condition but did not specify that sulfuric acid must be concentrated.

In question (b), the structure of benzocaine was drawn correctly.

In question (c)(i), the candidate named only one functional group out of three.

In question (c)(ii), the systematic name of benzocaine was written incorrectly.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

8 Benzocaine is a local anesthetic, which means that it makes only small areas of the body insensitive to pain. The preparation of benzocaine by the esterification of 4-aminobenzoic acid with ethanol is shown. $\text{H}_2\text{N}-\text{C}_6\text{H}_4-\text{COOH} + \text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{benzocaine} + \text{water}$ (a) State conditions for the reaction above. temperature 400° and catalyst [2] (b) Write the structural formula for benzocaine.  [1] (c) (i) Name all the functional groups in benzocaine. • carboxylic acid group [2] • amines group [2] (ii) Give the systematic name of benzocaine. 4-aminobenzoic acid [1]	For Examiner's Use 00 00 00 00
---	--

EXAMINER'S COMMENTS

The candidate received no marks.

In question (a), the candidate did not write the conditions correctly.

In question (b), the structure of benzocaine was drawn incorrect (it should be OCH_2CH_3).

In question (c)(i), the candidate named only one functional group.

In question (c)(ii), the systematic name of benzocaine was not written.

QUESTION 8(d) (i), (ii)

- (d) (i) Write down the structural formulae for the products of the reaction between benzocaine and hot aqueous sodium hydroxide.

[1]

- (ii) State the advantages of using sodium hydroxide in the hydrolysis of esters compared to using dilute acid.

.....

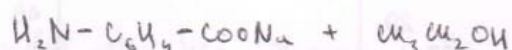
 [2]

MARK SCHEME

8 (d) (i)	$\text{H}_2\text{N}-\text{C}_6\text{H}_4-\text{COONa} + \text{CH}_3\text{CH}_2\text{OH}$	1	M1 for both correct products
8 (d) (ii)	The reactions are one-way rather than reversible	1	
	The products are easier to separate.	1	

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

- (d) (i) Write down the structural formulae for the products of the reaction between benzocaine and hot aqueous sodium hydroxide.



[1]

- (ii) State the advantages of using sodium hydroxide in the hydrolysis of esters compared to using dilute acid.

1) it is saponification reaction which is important for production purposes

2) reaction is not reversible only products form, no reversible reaction

[2]

3) less cost of production, no need of distill off the product.

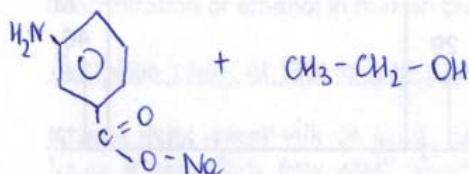
[Total: 9]

EXAMINER'S COMMENTS

The candidate received full marks for the correct products in (i) and for their ideas of the non-reversible reaction and easier separation in compared to acid hydrolysis.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

- (d) (i) Write down the structural formulae for the products of the reaction between benzocaine and hot aqueous sodium hydroxide.



[1]

- (ii) State the advantages of using sodium hydroxide in the hydrolysis of esters compared to using dilute acid.

1) Since alkaline hydrolysis is not reversible reaction like acid hydrolysis, it yields more products. 2) It (hydrolysis with NaOH) doesn't require high temp of 80°C for reaction to occur, so it's safer and less expensive. 3) It results in sodium salt, not carboxylic acid.

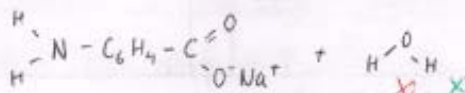
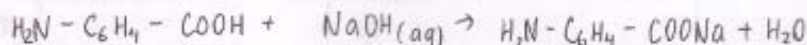
[2]

EXAMINER'S COMMENTS

The candidate received full marks for the correct products in (i) and for their ideas of non-reversible reaction in comparing with acid hydrolysis.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

- (d) (i) Write down the structural formulae for the products of the reaction between benzocaine and hot aqueous sodium hydroxide.



[1]

- (ii) State the advantages of using sodium hydroxide in the hydrolysis of esters compared to using dilute acid.

• NaOH, the rate of reaction is faster

• the yield of the reaction is higher

[2]

EXAMINER'S COMMENTS

The candidate received no marks. In question (ii), the candidate made a mistake with the second product.

In question (ii), the candidate did not use the key words.

QUESTION 9(a)(i), (ii), (iii)

9 Water dissociates to form hydroxonium ions and hydroxide ions.

Ex

(a) (i) Write down the equation that shows an acid-base reaction between two water molecules.

[1]

(ii) Complete the sentences to define the terms Bronsted-Lowry acid and Bronsted-Lowry base.

A Bronsted-Lowry acid is a

A Bronsted-Lowry base is a [2]

(iii) A Bronsted-Lowry acid has a conjugate base.

State the conjugate base of water.

[1]

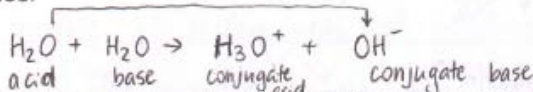
MARK SCHEME

9 (a) (i)	$\text{H}_2\text{O} + \text{H}_2\text{O} \leftrightarrow \text{H}_3\text{O}^+ + \text{OH}^-$	1	
9 (a) (ii)	Bronsted-Lowry acid is a proton donor Bronsted-Lowry base is a proton acceptor	1 1	
9 (a) (iii)	OH^- / hydroxide ion	1	

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

9 Water dissociates to form hydroxonium ions and hydroxide ions.

- (a) (i) Write down the equation that shows an acid-base reaction between two water molecules.



[1]

- (ii) Complete the sentences to define the terms Bronsted-Lowry acid and Bronsted-Lowry base.

A Bronsted-Lowry acid is a proton donor

A Bronsted-Lowry base is a proton acceptor

[2]

- (iii) A Bronsted-Lowry acid has a conjugate base.

H_3O^+ proton acceptor

State the conjugate base of water.

OH^-

[1]

is acid
electron
acceptor

For
Examiner's
Use

EXAMINER'S COMMENTS

The candidate received all four marks. The candidate correctly wrote the equation, completed the sentences about Bronsted-Lowry acid and base, and stated the conjugate base for water. In question (a)(i), a reference standard used in hydrogen NMR spectroscopy was identified correctly. In question (a)(ii), the given interpretation of peaks and absorption fully coincided with the mark scheme.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

9 Water dissociates to form hydroxonium ions and hydroxide ions.

- (a) (i) Write down the equation that shows an acid-base reaction between two water molecules.



[1]

- (ii) Complete the sentences to define the terms Bronsted-Lowry acid and Bronsted-Lowry base.

A Bronsted-Lowry acid is a H^+ donor

A Bronsted-Lowry base is a H^+ acceptor

[2]

- (iii) A Bronsted-Lowry acid has a conjugate base.

State the conjugate base of water.

OH^-

[1]

For
Examiner's
Use

EXAMINER'S COMMENTS

The candidate received three out of four marks.

In question (i), the equation was incorrect.

In question (ii), the candidate correctly explained what Bronsted-Lowry acid and base mean.

In question (iii), they correctly identified the conjugate base of water.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

9	Water dissociates to form hydroxonium ions and hydroxide ions.		For Examiner's Use
(a) (i)	Write down the equation that shows an acid-base reaction between two water molecules.	$\text{H}_2\text{O} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{OH}^-$	[1] 0
(ii)	Complete the sentences to define the terms Bronsted-Lowry acid and Bronsted-Lowry base.		
	A Bronsted-Lowry acid is a	proton acceptor	0
	A Bronsted-Lowry base is a	proton donor	0
(iii)	A Bronsted-Lowry acid has a conjugate base.		
	State the conjugate base of water.	$[\text{OH}]^{-2} + [\text{H}]^{+} \rightarrow [\text{OH}]^{-}$	1

EXAMINER'S COMMENTS

The candidate received only one mark out of four marks.
The candidate skipped the equation, incorrectly defined Bronsted-Lowry acid and base, and incorrectly identified the conjugate base of water.

QUESTION 9(b) (i), (ii)

(b) (i) Ethanoic acid, CH_3COOH , is a weak acid.

Write the equation for the dissociation of ethanoic acid.

[1]

(ii) Define the term a weak acid.

[1]

(c) The acid dissociation constant, K_a , for ethanoic acid is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$.

Write an expression for K_a .

Calculate both the $\text{p}K_a$ and the pH of $0.10 \times 10^{-3} \text{ mol dm}^{-3}$ ethanoic acid.

Give your answers to **two** decimal places.

$\text{p}K_a =$

$\text{pH} =$ [4]

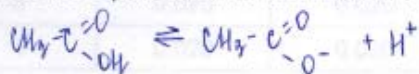
MARK SCHEME

9 (b) (i)	$\text{CH}_3\text{COOH} \leftrightarrow \text{CH}_3\text{COO}^- + \text{H}^+$ Or $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \leftrightarrow \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$	1	Do NOT accept without sign of equilibrium
9 (b) (ii)	partially/slightly dissociate in water and produces hydroxonium ions or free protons.	1	
9 (c)	$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$ $\text{p}K_a = -\log_{10}(K_a) = 4.74$ $[\text{H}^+] = \sqrt{1.8 \times 10^{-5} \times 0.10 \times 10^{-3}} = 4.24 \times 10^{-5}$ $\text{pH} = -\log [4.24 \times 10^{-5}] = 4.37$	1 1 1 1	Do NOT accept if an answer is not in two decimal places. Accept e.c.f.

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

- (b) (i) Ethanoic acid, CH_3COOH , is a weak acid.

Write the equation for the dissociation of ethanoic acid.



[1]

- (ii) Define the term a weak acid.

compound that doesn't fully dissociate into its conjugate base and H^+

[1]

- (c) The acid dissociation constant, K_a , for ethanoic acid is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$.

Write an expression for K_a .

Calculate both the $\text{p}K_a$ and the pH of $0.10 \times 10^{-3} \text{ mol dm}^{-3}$ ethanoic acid.

Give your answers to **two** decimal places.

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

$$K_a = \frac{[\text{H}^+]^2}{[\text{CH}_3\text{COOH}]} \quad (\text{for calculations})$$

$$[\text{H}^+] = \sqrt{K_a \cdot [\text{CH}_3\text{COOH}]} = \sqrt{1.8 \cdot 10^{-5} \cdot 0.10 \cdot 10^{-3}} = 4.24 \cdot 10^{-5} \text{ mol dm}^{-3}$$

$$\text{pH} = -\lg [\text{H}^+] = 4.37$$

$$\text{p}K_a = -\lg K_a = 4.74$$

$$\text{p}K_a = 4.74$$

$$\text{pH} = 4.37$$

[4]

EXAMINER'S COMMENTS

The candidate received all six marks.

In question (b)(i), the candidate wrote the equation correctly.

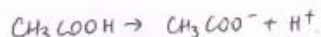
In question (b)(ii), the candidate defined the weak acid correctly.

In question (c), they calculated $\text{p}K_a$ and pH correctly.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

- (b) (i) Ethanoic acid, CH_3COOH , is a weak acid.

Write the equation for the dissociation of ethanoic acid.



[1]

- (ii) Define the term a weak acid.

Weak acid partially dissociates into ions.

[1]

- (c) The acid dissociation constant, K_a , for ethanoic acid is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$.

Write an expression for K_a .

Calculate both the $\text{p}K_a$ and the pH of $0.10 \times 10^{-3} \text{ mol dm}^{-3}$ ethanoic acid.

Give your answers to **two** decimal places.

1) $\text{p}K_a = -\log(K_a) = 4.74$

2) $K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$ 3) $\text{pH} = -\log(\text{H}^+) = 4.37$

$$[\text{H}^+]^2 = K_a \cdot 0.10 \times 10^{-3}$$

$$[\text{H}^+]^2 = 1.8 \times 10^{-5} \times 0.1 \times 10^{-3}$$

$$[\text{H}^+]^2 = 1.8 \times 10^{-9}$$

$$[\text{H}^+] = 4.24 \times 10^{-5}$$

$$\text{p}K_a = 4.74$$

$$\text{pH} = 4.37$$

[4]

EXAMINER'S COMMENTS

The candidate received five out of six marks.

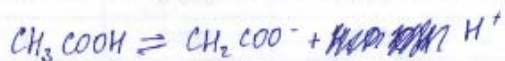
In question (b)(i), the candidate forgot to write the sign of the equilibria, but correctly wrote the description in question (b)(ii).

In question (c), the candidate correctly calculated $\text{p}K_a$ and pH, including two decimal places.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

- (b) (i) Ethanoic acid, CH_3COOH , is a weak acid.

Write the equation for the dissociation of ethanoic acid.



X

[1]

0

- (ii) Define the term a weak acid.

It is an acid which partially dissociates in a solution. [1]

1

- (c) The acid dissociation constant, K_a , for ethanoic acid is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$.

Write an expression for K_a .

Calculate both the $\text{p}K_a$ and the pH of $0.10 \times 10^{-3} \text{ mol dm}^{-3}$ ethanoic acid.

Give your answers to **two** decimal places.

$$K_a = \frac{[\text{CH}_2\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

✓

$$\text{pH} = -\log(4.243 \cdot 10^{-5}) = 4.4$$

XX

$$K_a = \frac{[\text{H}^+]^2}{[\text{CH}_3\text{COOH}]}$$

$$1.8 \cdot 10^{-5} = \frac{[\text{H}^+]^2}{0.10 \cdot 10^{-3}}$$

$$[\text{H}^+]^2 = 1.8 \cdot 10^{-5} \cdot 0.10 \cdot 10^{-3}$$

$$[\text{H}^+]^2 = 1.8 \cdot 10^{-9}$$

✓

$$\text{p}K_a = 4.2 \cdot 10^{-5}$$

$$\text{pH} = 4.4$$

[4]

2

EXAMINER'S COMMENTS

The candidate received three out of six marks.

In question (b) (i), the candidate wrote the incorrect formula for the acetate ion but correctly wrote the description in question (b)(ii).

In question (c), the candidate correctly calculated the pH but forgot to write it to two decimal places and did not calculate the $\text{p}K_a$.

QUESTION 9(d)

(d) Calculate the pH of a solution of 0.10 mol dm⁻³ sulfuric acid.

Give your answers to **two** decimal places.

E

[3]

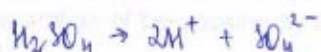
MARK SCHEME

9 (d)	$\text{H}_2\text{SO}_4 \rightarrow 2\text{H}^+ + \text{SO}_4^{2-}$	1	Give three marks if answer is correct with two decimal places
	$[\text{H}^+] = 2 \times 0.10 = 0.2 \text{ mol dm}^{-3}$	1	Give two marks if answer is 1.00 and $\text{H}_2\text{SO}_4 \rightarrow \text{H}^+ + \text{HSO}_4^-$
	$\text{pH} = -\log [0.2] = 0.70$	1	Give one mark if answer is 1.30

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

(d) Calculate the pH of a solution of 0.10 mol dm⁻³ sulfuric acid.

Give your answers to **two** decimal places.



$$[\text{H}^+] = 2[\text{H}_2\text{SO}_4] = 2 \times 0.10 = 0.20 \text{ mol dm}^{-3}$$

$$\text{pH} = -\lg[\text{H}^+] = 0.70$$

Answer: pH = 0.70

[3]

For
Examiner's
Use

3

EXAMINER'S COMMENTS

The candidate received all three marks.

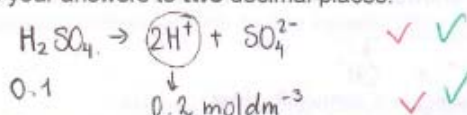
The candidate wrote the equation, calculated the concentration of the hydrogen ion, and determined the pH of the sulfuric acid.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

(d) Calculate the pH of a solution of 0.10 mol dm^{-3} sulfuric acid.

For
Examiner's
Use

Give your answers to two decimal places.



$$\text{pH} = -\log(\text{H}^+) = -\log(0.2) = 0.7$$

[3]

2.2

EXAMINER'S COMMENTS

The candidate received only two marks out of three.

The candidate performed all calculations correctly but did not provide the answer to two decimal places.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

(d) Calculate the pH of a solution of 0.10 mol dm^{-3} sulfuric acid.

For
Examiner's
Use

Give your answers to two decimal places.

$$[\text{H}^+]^2 = 1.8 \cdot 10^{-5} \cdot 0.10$$

$$[\text{H}^+]^2 = 1.8 \cdot 10^{-6}$$

$$[\text{H}^+] = 1.342 \cdot 10^{-3}$$

$$\text{pH} = -\log 1.342 \cdot 10^{-3} = 2.9$$

$$\text{pH} = 2.9$$

[3]

[Total 13]

XXX

0

(5)

EXAMINER'S COMMENTS

The candidate received no marks.

The candidate attempted to calculate the pH of the sulfuric acid as a weak acid but did not provide the answer to two decimal places.

QUESTION 10(a)

10 Reaction kinetics is the study of the rates of chemical reactions.

Ex.

(a) Explain the effect of the surface area of reactants on reaction rates.

.....

.....

.....

..... [3]

MARK SCHEME

10 (a)	the powdered solid has a greater surface area	1	
	the faster the rate of reaction	1	
	because more collisions will occur	1	
		[3]	

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

10 Reaction kinetics is the study of the rates of chemical reactions.

For
Examiner's
Use

(a) Explain the effect of the surface area of reactants on reaction rates.

The increase in surface area of reactants increase the reaction rate. The increasing surface area means more areas for collisions of reactants which increase the rate and the production of products.

[3]

3 3

EXAMINER'S COMMENTS

The candidate received all three marks.

The candidate clearly showed all the key words in the answer.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

10 Reaction kinetics is the study of the rates of chemical reactions.

For
Examiner's
Use

(a) Explain the effect of the surface area of reactants on reaction rates.

The increase of surface area of reactants speeds up the reaction, because there is more place for reactants to react and ~~react~~ form products. Therefore, the surface area's increase rises the rate of reaction.

[3]

2 2

EXAMINER'S COMMENTS

The candidate received two out of three marks.

The candidate stated how an increase in surface area affects the rate of reaction but did not explain it using the collision theory.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

10 Reaction kinetics is the study of the rates of chemical reactions.

For
Examiner's
Use

(a) Explain the effect of the surface area of reactants on reaction rates.

the smaller the surface area, the higher number of the successful collisions occur, because more particles undergo the reaction, so the reaction rate increases.

[3]

0 0

EXAMINER'S COMMENTS

The candidate received no marks.

The candidate incorrectly wrote that a smaller surface leads to more collisions and an increase in reaction rate, which is an incorrect idea and contradictory.

QUESTION 10 (b) (i), (ii), (iii)

(b) The results of rate measurements of the reaction between A, B, and C compounds are given in the table below.

Experiment	Concentration / mol dm ⁻³			Rate of loss of A / mol dm ⁻³ s ⁻¹
	A	B	C	
1	0.010	0.020	0.005	1.4×10^{-6}
2	0.010	0.010	0.005	7.0×10^{-7}
3	0.020	0.020	0.005	1.4×10^{-6}
4	0.020	0.020	0.015	4.2×10^{-6}

(i) Deduce the order of reaction with respect to A, B and C, and the overall order of the reaction.

order of reaction with respect to A

order of reaction with respect to B

order of reaction with respect to C

overall order of reaction [3]

(ii) Write the rate equation for the reaction.

..... [1]

(iii) Calculate a value for the rate constant including units.

rate constant = unit [2]

MARK SCHEME

10 (b) (i)	Two marks: Order of reaction with respect to A - 0 Order of reaction with respect to B - 1 Order of reaction with respect to C - 1 One mark: Overall order of reaction = 2	2 1 [3]	M2 – three correct answers M1 – two correct answers
10 (b) (ii)	Rate = k [B][C]	[1]	ecf
10 (b) (iii)	k = 0.014 mol ⁻¹ dm ³ s ⁻¹	1 1 [2]	ecf

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

order of reaction with respect to A 0

order of reaction with respect to B 1 ✓

order of reaction with respect to C 1 ✓

overall order of reaction 2 ✓ [3] **3**

(ii) Write the rate equation for the reaction.

rate = k [B] [C] ✓ [1] **1**

(iii) Calculate a value for the rate constant including units.

$$k = \frac{\text{rate}}{[B][C]} = \frac{1.4 \times 10^{-6}}{0.020 \times 0.005} = 0.014 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$$

rate constant = 0.014 ✓ unit mol⁻¹ dm³ s⁻¹ ✓ [2] **2**

EXAMINER'S COMMENTS

The candidate received all six marks.

The candidate correctly identified the orders of the reaction, wrote the rate equation, and calculated the rate constant with units.

SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

(i) Deduce the order of reaction with respect to A, B and C, and the overall order of the reaction.

rate 1 = $k[A]^m[B]^n[C]^p$ order of reaction with respect to A 0 ✓ ✓

rate 2 = $k[A]^m[B]^n[C]^p$ order of reaction with respect to B 2 ✗ ✗

$\frac{1.4 \times 10^{-6}}{7 \times 10^{-7}} = \frac{(0.02)^m}{(0.01)^m}$ order of reaction with respect to C 1 ✓ ✓

$2 \times 10^{-4} = (2 \times 10^{-3})^n$ overall order of reaction 3 ✗ ✗ [3] **1 1**

(ii) Write the rate equation for the reaction.

rate = k [A]⁰ [B]² [C]¹ ✓ ✓ ✓ [1] **1 1**

(iii) Calculate a value for the rate constant including units.

$$k = \frac{4.2 \times 10^{-6} \text{ mol dm}^{-3} \text{ s}^{-1}}{(0.02)^2 \times (0.015)} = \frac{4.2 \times 10^{-6}}{6 \times 10^{-6}} = 0.7$$

rate constant = 0.7 ✗ ✓ unit mol⁻² dm⁶ s⁻¹ ✗ ✓ [2] **0 2**

EXAMINER'S COMMENTS

The candidate received four marks out of six.

In question (i), the order of B and the overall order is incorrect.

In questions (ii) and (iii), an error was carried forward, but all procedures were correct except for incorrect numbers.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

(i) Deduce the order of reaction with respect to A, B and C, and the overall order of the reaction.

order of reaction with respect to A	first (1 st)	X	
order of reaction with respect to B	second (2 nd)	X	
order of reaction with respect to C	third (3 rd)	X	
overall order of reaction	third (3 rd)	X [3]	0

(ii) Write the rate equation for the reaction.

~~$k[A][B]^2[C]^3$~~ $\frac{k[A][B]^2}{[C]^3}$ X [1] 0

(iii) Calculate a value for the rate constant including units.

$$\frac{[A][B]^2}{[C]^3} = \frac{\text{mol dm}^{-3} \text{s}^{-1} \cdot \text{mol}^2 \text{dm}^{-6} \text{s}^{-2}}{\text{mol}^3 \text{dm}^{-9} \text{s}^{-3}} =$$

$$\frac{0,010 \cdot (0,020)^2}{(0,005)^3} = \frac{4 \cdot 10^{-6}}{(0,005)^3} = 32$$

rate constant = 32 unit $\text{mol}^6 \text{dm}^{-1} \text{s}^{-6}$ [2] 0

$(4 \cdot 10^{-6}) \cdot (0,05)^3 = 4 \cdot 10^{-6}$

[Total 9] (2)

EXAMINER'S COMMENTS

The candidate received no mark.

The candidate incorrectly identified the orders of the reaction in question (i), wrote the rate equation incorrectly in question (ii), and miscalculated the rate constant.

QUESTION 11 (a) (b) (i), (ii)

11 Amines are derivatives of ammonia.

A primary amine can be prepared from the reaction of 2-bromo-2-methylpropane and ammonia.

(a) Write a balanced chemical equation for this reaction.

All organic compounds should be shown as their structural formulae.

..... [1]

(b) (i) Explain why ammonia should be in excess in the reaction.

.....

 [1]

(ii) State the other reaction conditions.

..... [1]

MARK SCHEME

11 (a)	$(\text{CH}_3)_3\text{CNH}_2 + \text{NH}_4^+\text{Br}^- / (\text{CH}_3)_3\text{CNH}_2 + \text{HBr}$	[1]	Organic compounds should be in their structural formulas.
11 (b) (i)	It ensures that all the halogenoalkane molecules react with the ammonia to produce a primary amine.	1	
11 (b) (ii)	a sealed tube / heating / ethanol as a solvent/ alcoholic solvent	1	

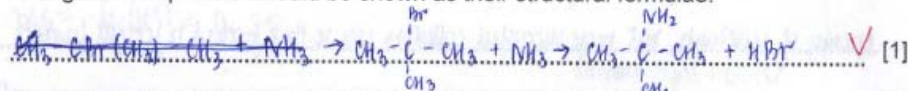
SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

11 Amines are derivatives of ammonia.

A primary amine can be prepared from the reaction of 2-bromo-2-methylpropane and ammonia.

(a) Write a balanced chemical equation for this reaction.

All organic compounds should be shown as their structural formulae.



(b) (i) Explain why ammonia should be in **excess** in the reaction.

Ammonia should be in excess to prevent formation of secondary/tertiary amines because if haloalkane is in excess, primary amine will be formed in low yield.

(ii) State the **other** reaction conditions.

heat

For
Examiner's
Use

EXAMINER'S COMMENTS

The candidate received all three marks.

In question (a), the candidate wrote the equation correctly, including the structural formulae.

In question (b), the candidate showed a clear idea of the formation of only a primary amine and the conditions required.

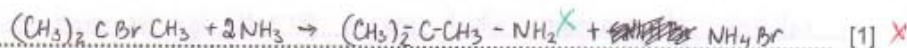
SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

11 Amines are derivatives of ammonia.

A primary amine can be prepared from the reaction of 2-bromo-2-methylpropane and ammonia.

(a) Write a balanced chemical equation for this reaction.

All organic compounds should be shown as their structural formulae.



(b) (i) Explain why ammonia should be in **excess** in the reaction.

First ammonia replaces the bromine atom, and then second ammonia attach to it a hydrogen atom to form amine, towards itself a hydrogen atom from the attached amino group to form NH_2 .

(ii) State the **other** reaction conditions.

NH_3 in ethanol

For
Examiner's
Use

EXAMINER'S COMMENTS

The candidate received one out of three marks.

In question (a)(i), the candidate did not write the correct equation.

In question (b)(i), the candidate did not include the key words. However, in question (b)(ii), the candidate wrote the correct condition.

SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

16

11 Amines are derivatives of ammonia.

A primary amine can be prepared from the reaction of 2-bromo-2-methylpropane and ammonia.

(a) Write a balanced chemical equation for this reaction.

All organic compounds should be shown as their structural formulae.

$$\text{NH}_3 + \text{H}-\text{C}(\text{CH}_3)_2-\text{Br} \rightarrow \text{H}-\text{C}(\text{CH}_3)_2-\text{NH}_2 + \text{H}_2\text{Br}$$

(b) (i) Explain why ammonia should be in **excess** in the reaction.

Because there are delocalized electrons that should be substituted with large surface area.

(ii) State the **other** reaction conditions.

temperature.

For Examiner Use

EXAMINER'S COMMENTS

The candidate received no marks.

The candidate did not write the correct equation in question (a)(i).

In question (b)(i), the candidate did not answer the question. However, in (b)(ii), the candidate wrote 'temperature' instead of 'heating,' which has a different meaning.

QUESTION 11(c) (d)

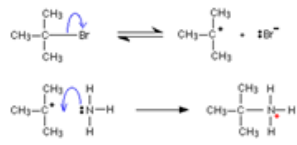
(c) Give the name of this type of reaction.

..... [1]

(d) Draw the mechanism for the reaction.

[3]

MARK SCHEME

11 (c)	nucleophilic substitution reaction/ SN	[1]	Do NOT accept if only substitution reaction is given
11 (d)	 A curly arrow from the C-Br bond to the bromine atom A curly arrow from the lone pair of ammonia to the (CH)₃C⁺ carbocation For all correct charges and lone pairs (on nitrogen and bromide ion)	1 1 1 [3]	Credit only for all correct charges and lone pairs (on nitrogen and bromide ion)

SAMPLE OF CANDIDATE RESPONSE GRADED 'A'

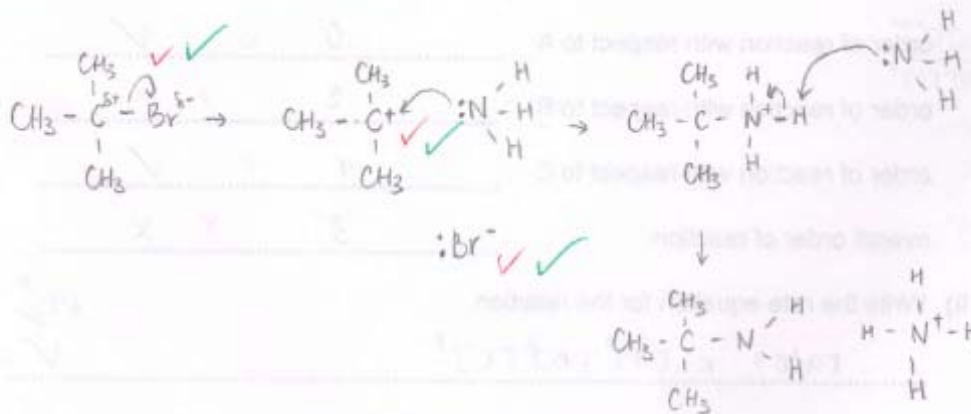
(c) Give the name of this type of reaction.

.....nucleophilic..... substitution (SN₁)

[1] ✓

1 1

(d) Draw the mechanism for the reaction.



[3]

3 3

EXAMINER'S COMMENTS

The candidate received all five marks. In question (c), correctly identified the mechanism and drew it in question (d).

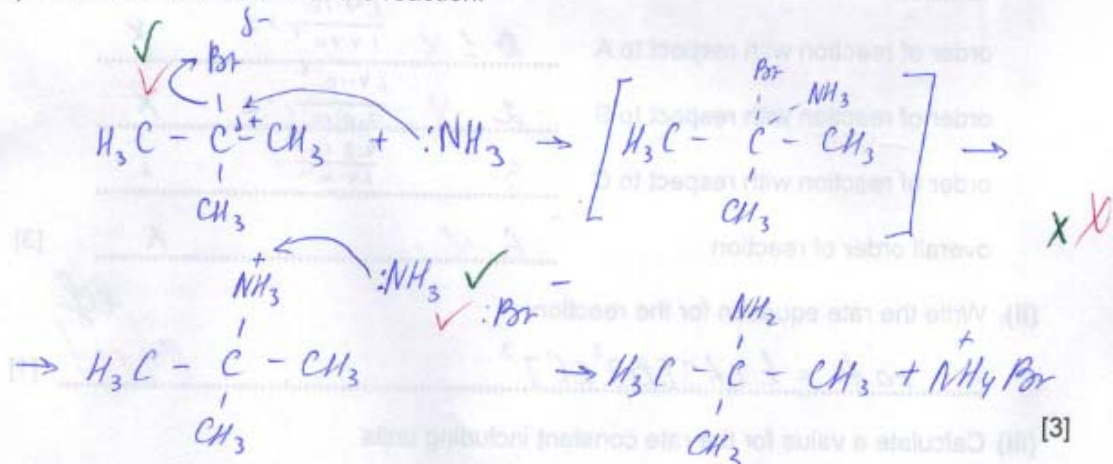
SAMPLE OF CANDIDATE RESPONSE GRADED 'C'

(c) Give the name of this type of reaction.

Nucleophilic substitution / SN2

[1] ✓ 1 1

(d) Draw the mechanism for the reaction.



[3] 2 2

EXAMINER'S COMMENTS

The candidate received three out of four marks.

In question (c), correctly named the type of reaction.

In question (d), the candidate showed a curly arrow from the CBr bond to bromine and the lone pair on the bromine ion, but they incorrectly drew the intermediate step as SN1.

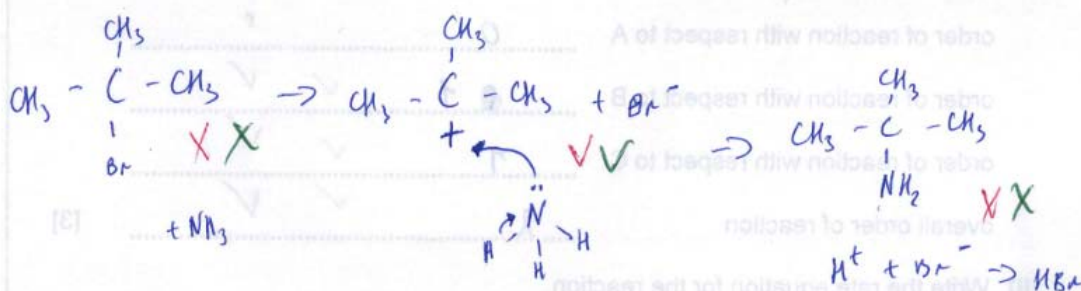
SAMPLE OF CANDIDATE RESPONSE GRADED 'E'

(c) Give the name of this type of reaction.

nucleophilic substitution

[1] ✓ ✓

(d) Draw the mechanism for the reaction.



[3]

EXAMINER'S COMMENTS

The candidate received two out of four marks.

In question (c), the candidate correctly named the type of reaction.

In question (d), the candidate did not show a curly arrow from the CBr bond to the bromine and the lone pair on the bromine ion. However, the candidate correctly showed the intermediate SN2 step.